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ENERGY - HOW TO PRODUCE IT WITHOUT DESTROYING THE HOME OF OUR CHILDREN

"There is no crisis for energy, there is a crisis of ignorance" B. Fuller

- PAGE 1 -

BEFORE YOU BEGIN

After moving to this site (www.quanthomme.org) Mr. Pantone 's plans , the demonstrations and the experience we gained and their detailed descriptions, there are still many questions to be clarified on the functioning of this system.

Since there are many people who want to equip different types of bikes with this device, I decided that I can help by sharing with you my my humble experience in this regard.

I guess you are not a complete beginner on this topic and you are interested in everything we have done and described about the little ones. four-stroke engines, which we converted to alternative fuel. You could find that the principle of Mr. GEET reactor PANTONE is damn simple that its operation remains a bit unclear so far, the explanations given by different people are contradictory.

This does not matter, as the device works, is easy to make and applicable to all types of motors.

I remind you that the patented device consists of a simple iron tube, in the middle of which the axis is fixed steel rod with a diameter sufficient to leave a distance of 1-2 mm between the rod and the walls of the pipe. For the efficient operation of the device you need to provide heating of this tube containing the steel rod. This heating is done by installing the reactor tube in the inside of another metal pipe in which the exhaust gases circulate (see diagram).

When the evaporated particles of any hydrocarbon pass through the reactor tube, they decompose into combustible hydrogen gas. The efficiency of this reaction is improved by increasing the reactor temperature and the magnetic state of the North / South oriented steel rod in the first development.



No matter what actually happens in the reactor, the fact is that it produces gas of such quality that it works with any four-stroke engine with no matter what the level of compression, with negligible pollution, and gives significant fuel savings without affecting power.

WHAT WE NEED TO KNOW BEFORE STARTING CONSTRUCTION AND INSTALLATION OF GEET PANTONE IN ENGINE

All the explanations below are intended to save you time and show you the difficulties you need to overcome. If from the very beginning you failed to do properly the evaporator of the selected fuel, you will not be able to activate

and manage your GEET reactor well.

It should be clear to you that it is better for fuel vapor to pass through the reactor tube than for sprayed fuel. If you still try to if you run sprayed fuel through the reactor, it should be sprayed as finely as possible.

From the beginning of your project, you need to choose the fuel and the way it will pass through the reactor - evaporation or spraying. Whatever and the choice to make, you will encounter a lot of difficulties in doing the right thing on one or the other system. My comments on these two systems will help you in your choice.

There is another difficulty that occurs with all types of engines - at all engine speeds to deliver optimal proportional mixture of GEET gas and air, which is necessary for optimal combustion and purity of the exhaust gases.

In principle, you would not have any particular problems coming from the reactor itself, because it will work immediately if it is done exactly as planned.

Some of the manufacturing problems will come from the evaporation or atomization system of the fuel used, and others Difficulties come in installing and adjusting the system of valves designed to power the motor in any mode.

So - to supply fuel through the reactor, you have a choice between two options: 1) spraying; 2) evaporation

1) Spraying

It should be as fine as possible. It can be realized with the help of a system of injector and high pressure pump or from something like a carburetor or a compressed air atomizer, like paint guns.

I started experimenting with these, they work, but they are very difficult to set up and do not spray enough on some fuels.

I will leave aside the comments on the spraying for now and return to them when I have achieved some satisfactory results in my experiments, as it has some advantages over evaporation that are not to be overlooked.

2) Evaporation

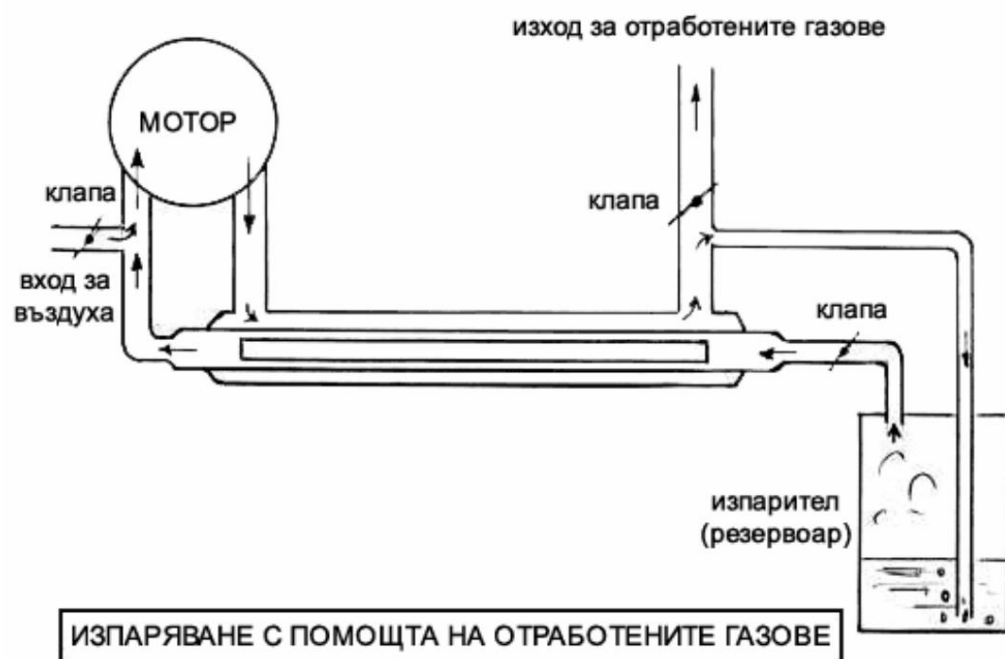
It is used in all installations shown and is the easiest option to implement in experimental installations designed for acquaintance with the invention of Mr. PANTONE, especially when we use as a fuel easily volatile at low temperatures product, such as gasoline. But we should not be deceived, this option is also difficult to master and refine when we want to apply it seriously in practice on a car engine or other motorcycle designed for everyday use.

With the evaporation method, the problems associated with mixing the air with the gas coming out of the reactor are as delicate to solve as with the spray.

Conversely, evaporation seems to give better results in preparing the fuel for the process that takes place in the GEET reactor.

You may also have some problems with the settings for maintaining a constant amount of fuel in the evaporator (tank), especially when you decide to add water.

That is, for our first attempts, let's choose evaporation as a method and gasoline as a fuel, and possibly add water. Gasoline is evaporates easily at low temperatures, it is enough just to pass through it (to be dispersed) with air or some gas, so it is more interesting to work with it.



First of all, let's remember again that we should not be confused evaporation and atomization. The evaporation that interests us here, is caused by the passage of air or burnt gases through the fuel mixture, slightly heated for convenience.

Evaporation by heating the fuel is not recommended because for some fuels the reactor does not work or works poorly. So it should be avoided when we use the exhaust gases for the evaporator, not to rise the temperature until the fuel boils, but be accurate as long as it is sufficient to facilitate evaporation.

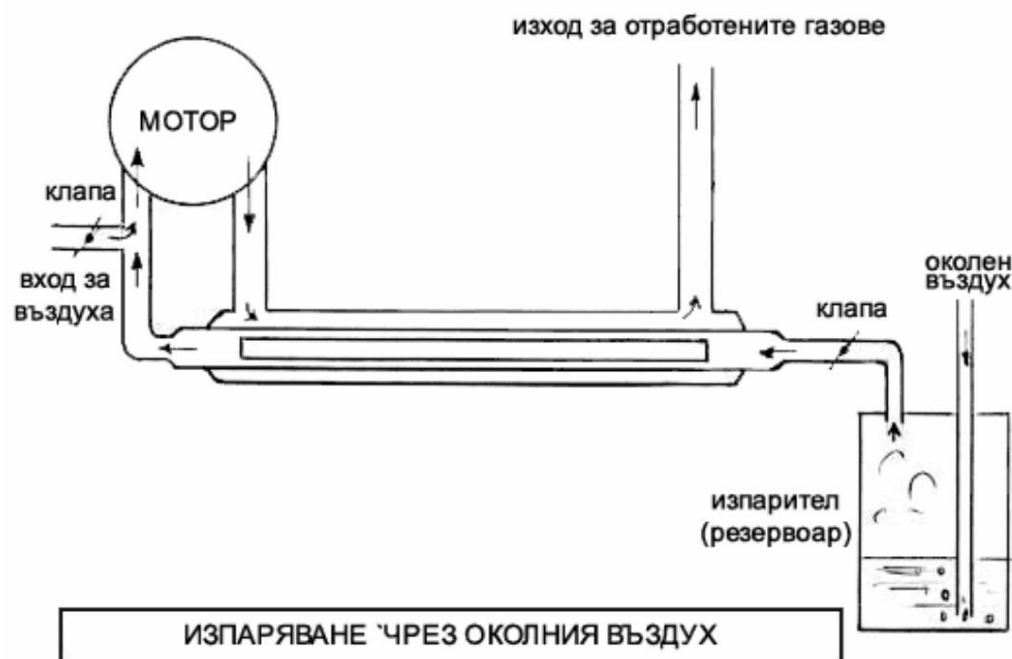
With the evaporator you have the opportunity to evaporate the fuel or with using ambient air, or using exhaust gases. The second option is applied in all installations described so far.

(a) Exhaust evaporation.

This way is to skip some of the burnt gases through the fuel mixture in the vaporizer (tank), which causes "fuming", the fuel vapor mixed with some of the exhaust gases fill the top of the evaporator (tank) before being sucked into the reactor.

This simple but easy-to-try solution has more disadvantages than advantages. In this case, the heat of the exhaust gases prevents the fuel from cooling down and stop evaporating enough - this seems to be an advantage, but in practice the temperature in the evaporator will be unsafe, evaporation can vary between insufficient and too much, which will make it difficult to control the engine,

which will be possible only by constantly adjusting the valves. Thus, they cannot be achieved at any engine speed the exact proportions in the fuel mixture. In conclusion, if we try to evaporate fuel with uneven temperature with gas, which too varies its temperature, we will never achieve a uniform linear power supply to the reactor and we will have to constantly adjust and we change the different valves without being sure at which point it works optimally.



In case you are "evaporating" with the help of the exhaust gases, you should be well aware that there is no way to know if the money is "pushed" into the reactor or "sucked in". In time I noticed that we need to keep it light overpressure in the evaporator (tank) and adjust the flow of money in the direction of the reactor with an additional valve absolutely necessary, which is located between the outlet of the reactor and the inlet to the engine, as simultaneous adjusting these two valves gives the exact proportions of fuel gas mixture for the engine. All this is quite complicated and it is not easy to implement.

With this option we also have the following problem - the evaporator, which is under some pressure, especially when gasoline is mixed with water (need for more intensive evaporation for gasoline) we find that the money coming out of the evaporator is more "ticked" in the reactor than "sucked". This leads until a large part of the gas leaving the reactor is discharged through the air valve when the engine is not in the charging period (?) (25% of the time) (?). This problem mainly concerns

engines with less than 4 cylinders. To eliminate this inconvenience, a dispositive device can be placed in the valve, which interferes with return or find a way to keep the reactor always depressed, which in our case is practically impossible.

In short, if you have experimented with the experimental reactor and its four valves, you have probably already found out for yourself what I explained above, so before you start with the actual installation, you need to learn some details.

b) Evaporation by ambient air.

Instead of causing the fuel to evaporate by injecting some of the flue gases into the evaporator, with all its benefits and disadvantages described above, we can be content to use the ambient air for this purpose.

In this case, we use an ordinary pipe that enters at one end of the evaporator and the fuel and is not connected to anything else. The depression caused in the evaporator (tank) by the suction of the motor will give the necessary evaporation.

This system works well, but it is difficult to reduce the speed of the motor due to poor suction at slow stroke (when it comes to small bikes). Conversely, this method can use something like a gas carburetor, which with a single valve will maintain the required proportions in the gas fuel mixture at any engine speed (see diagram).

There is one inconvenience that is easy to overcome - a few moments after starting the gasoline cools and stops evaporating enough to cause the engine to stop. In the first method, the exhaust gases compensate with their temperature for this cooling of fuel. As it is very easy to make an evaporator (tank) with a constant temperature, this inconvenience is not an obstacle, but even an advantage, as we can have an evaporator regulated at a constant temperature depending on the choice of fuel. It is even possible, if necessary, to heat to the required temperature the air entering the evaporator, which is an interesting solution for less volatile fuel.

Once you get acquainted with these two evaporation systems, you will understand that only the second allows a simple and inexpensive installation, easy settings, which you will find out yourself after reading the following descriptions. This system allows a good functioning on the engine with only one valve, which is actually a simple carburetor, slightly modified for this purpose.

FOR THE USE OF WATER IN AN ENGINE EQUIPPED WITH A GEET REACTOR

PANTONE

All experiments with engines equipped with GEET show that a significant percentage of water can be added to gasoline and "consumed" by the engine, having previously very likely converted to hydrogen and oxygen in the reactor, but under certain conditions.

If we add water to the gasoline in the evaporator, it will partially evaporate together with the gasoline due to the incoming air or exhaust gases passing through the evaporator, depending on the selected system, the water will decompose in the reactor of hydrogen and oxygen and these gases will be added to the gases produced by the decomposition of petrol (probably hydrogen and carbon or carbon monoxide).

To your knowledge, keep in mind that when water and gasoline are used in the same evaporator, there are certain limits to good operation in which if the evaporator is too cold, the petrol will evaporate more than the water, the same will happen when the temperature is higher - gasoline will evaporate more than water because it is more volatile than it. The lower limit of good functioning is critical. It will be better to install two evaporators to control the situation, to be heated on the required temperature to maintain the required evaporation. I haven't experimented with two evaporators yet, but I'll try soon. The cold engine will run on gasoline and little by little, with the heating of the evaporator containing water, water vapor will be added to petrol. In addition, it is easier to fill and maintain two separate evaporators, for water and petrol, using tanks with level indicators (arrow float). Do not forget to add alcohol to the water in cold weather.

Problems with water use make some people disappointed on their first try. The processor was not introduced by its inventor as a water engine, but as a device that can consume water under certain conditions. So we have to have these conditions face.

SOME NOTES THAT MAY BE USEFUL TO YOU.

When you make your first attempts in cold weather, you will notice that the next day the engine starts hard with the gasoline left in the evaporator from yesterday, while before your engine worked very well, once warmed up. You will only notice this problem if you choose the second system - evaporation from ambient air. This is due to the fact that the gasoline left in the evaporator from the previous day has been lost its most volatile part and that when the weather is cold, ordinary evaporation by suction of ambient air cannot cause enough gasoline fumes to run the engine. For a simple experience, you can be content to supplement the evaporator with fresh gasoline to the required level, but for a serious practical application a more serious solution to this inconvenience must be found, for to feed the evaporator constantly with fresh gasoline. In this case, a manual or automatic vacuum cleaner will have to be installed with a cold evaporator that will only send the exhaust gases through the evaporator for a short time, enough to cause warming and evaporation. This vacuum cleaner is easy to make, either using a manual valve or a thermostatic valve, as for heating the evaporator. Thus, after a short time at the moment when the evaporator is heated to the required temperature, the vacuum cleaner is switched off.

I have already explained to you that I have not yet tried to do an installation with two evaporators, separate for water and petrol, but if you are progressing faster than me, construct two identical evaporators (tanks) with a heating coil and a thermostatic valve.

For heating coils, copper or stainless steel pipes, long enough, 10 or 12 mm in diameter. You need to ensure in sufficient dimensions, because if the gasoline evaporator needs only 25-30 degrees, the water evaporator will need heating between 60 and 80 degrees. If you fail to reach the temperature required for the evaporator, using some of the exhaust gases at the outlet of reactor, drill directly to the exhaust at the reactor inlet.

If you want to improve your design, you can provide heating to a constant temperature of the air that maintains evaporation. just before it enters the evaporator. You can easily fix this rather unstable parameter. Do it another small heat exchanger with a small tank equipped with a coil and a thermostatic valve, as in the evaporator, install more a coil through which air will circulate. Fill this heat exchanger with some liquid before putting it into operation. With this device, the air introduced into the evaporator and causing evaporation will maintain the desired temperature after starting.

It would also be interesting to insulate the reactor with a "vest" of glass wool or other thermal insulation product to maintains high temperature in all conditions.

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EXPERIMENT WITH FOUR-STROKE ENGINE 144 CM3 WITH CARBURETOR

FEBRUARY 21, 2001

I will now describe in as much detail as possible my experiment as an improved solution on a small four-stroke Honda 144 cm3 motor.

[Click on the photo to see more explanations, photos and videos](#)



Honda 144 cm3 engine, the second modified engine.

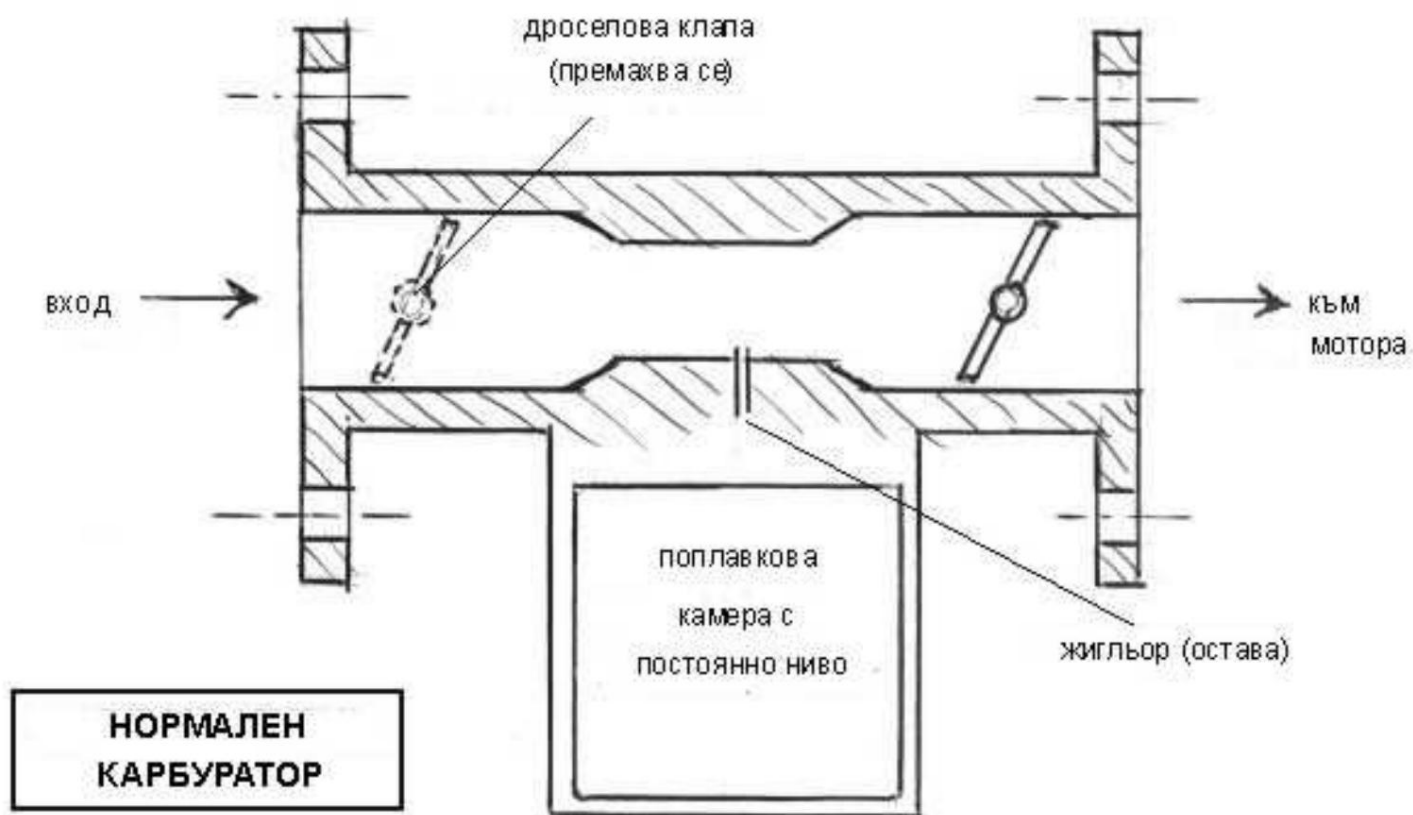
- For fuel I chose gasoline without adding water.
- I decided to control the motor with one valve (modified arboretum).
- I chose to use the ambient air to ensure the vaporization of gasoline.
- I made an evaporator heated by the exhaust gases with a thermostatic control valve.
- I chose a "cold start" system to cause the fuel to evaporate.

The use of the method of inducing the evaporation of fuel by suction of ambient air and its passage through the fuel, the choice of evaporator heated to a constant temperature and maintaining a constant level of fuel in the evaporator, as in carburetors, allows to fix the most important parameters. It remains to make a device to heat the air to constant temperature, but this will be for the near future.

The choice of the above methods and means allows to control the engine with a single valve, or more precisely with the modified carburetor, which can still function as a normal carburetor if necessary.

MODIFIED CARBURETOR

Take the original carburetor of the engine, remove the air filter and the starter cover (valve?) Which is located at the inlet (see diagram below).



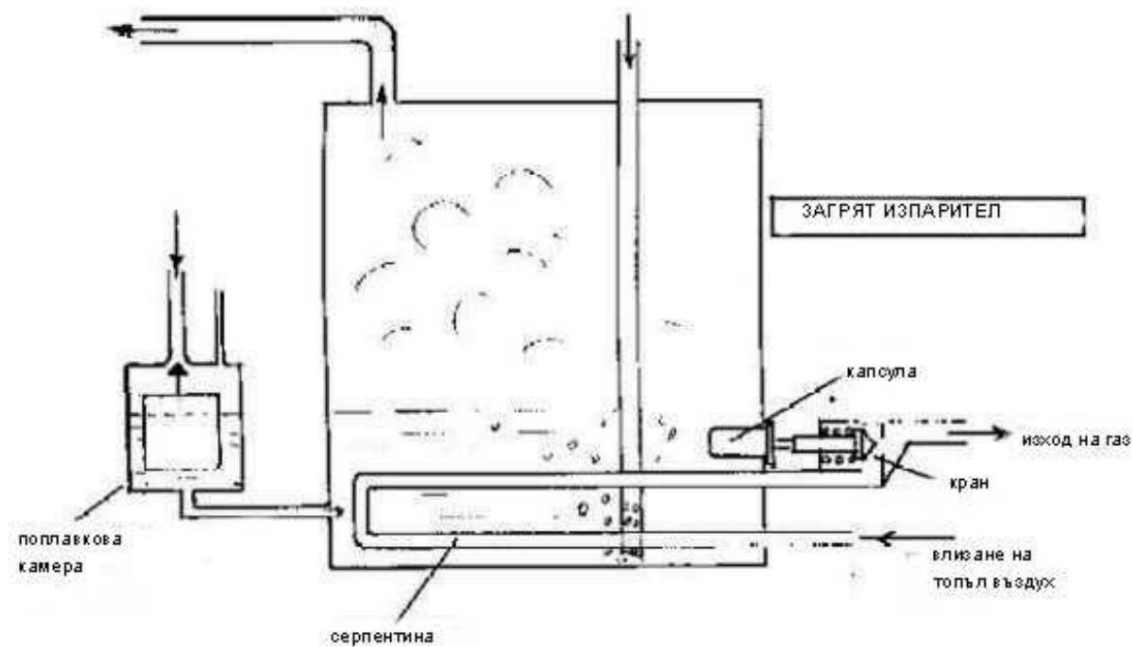
Keep everything else in the carburetor, which can be used later for start-up or for future experiments with fuels such as fuel oil or other products in need of pre-heating of the entire system. An adjustable nozzle must be fitted to the carburetor to suck the gas leaving the reactor (see diagram). This one The jig consists of a bushing mounted in the center of the carburetor nozzle and having a conical part whose position can be changed to regulate the amount of gas in relation to the air. This regulation will be different according to the fuel used, depending on whether it is mixed with water and in what proportions. The space between the sleeve and the nozzle should be quite narrow and proportional to engine cylinders to obtain a nozzle depression sufficient to maintain a minimum of evaporation at low operating conditions. It would be best for this space to be able to vary according to the throttle opening, which leads to the creation of a special carburetor with varying nozzle. Later I will give you the plan of such a carburetor, which I invented myself. Pay attention to the position at the end of the nozzle bushing with respect to the narrowed part of the nozzle, you must adjust it by testing, loosening the bushing and trying different forward or backward / backward positions.

Of course, you can find other solutions for the formation of the fuel mixture, harness your imagination. What I describe to you, it is not complicated at all, you can improve it if you have good knowledge of engines, fuels and chemistry.

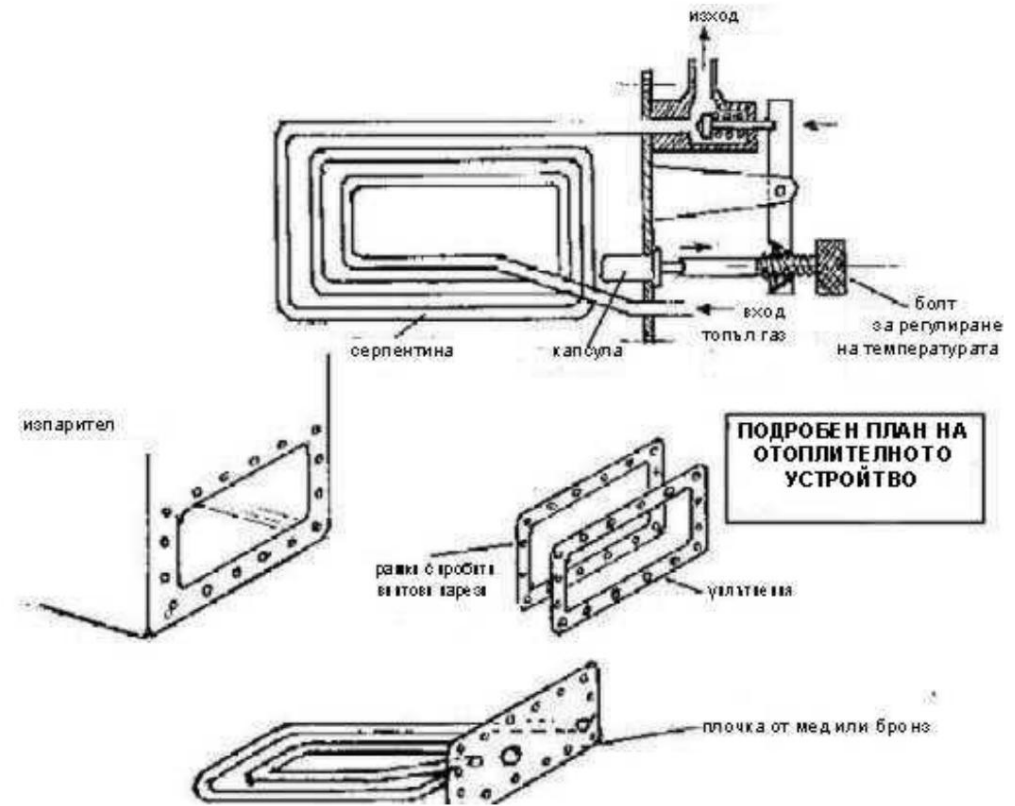
To start cold, you can install a rotating air valve at the air inlet in the carburetor, and through this inlet will also spray some gasoline with a spray bottle, which is easier to start than to fill the carburetor.

EVAPORATOR WITH ADJUSTABLE HEATING

The heating of the evaporator is provided by the passage of exhaust gases, some of which are intercepted at the outlet of the reactor and we direct to a metal coil mounted at the bottom of the evaporator. The regulation of the amount of hot gases in the coil is controlled by a thermostatic valve from a radiator for central heating.



In general, the device is easy to make, with a little dexterity or with the help of a friend-connoisseur. Remove the thermostat capsule and remove it separate from the tap. Examine the diagrams and make a similar construction, which you will then install in the evaporator after you have cut the necessary hole in it. The diagrams are shown much better than in a long description.



SOME IMPORTANT THINGS FOR FUEL BURNING IN ENGINES

What is written below does not apply to those for whom internal combustion engines have no secrets, but to those who are a little more beginner than me on this topic, but who also want to try their hand at the practical application of GEET technology.

GASOLINE ENGINE (moteur à explosion)

Let's remember - in order for an engine to function well and pollute less, the fuel used must burn completely in appropriate conditions. In a gasoline engine, or also called an internal combustion engine, gasoline enters the cylinder in the form of droplets mixed with air. These droplets are created and sprayed by a carburetor or injection system.

Inside the cylinder of the petrol engine, each of these droplets represents a huge sphere compared to the surrounding air molecules on which only the surface is in contact with the air, which contains the oxygen needed for combustion. Currently on ignition, initially only the surface of the sphere is ignited, and not even the whole, because the air contains a large

part of non-combustible gases (80% nitrogen, for example), which interferes to some extent with combustion. Of course, the burning still continues, as it should be deterred to some extent by the creation of exhaust gases. Despite these obstacles, combustion seems to be a very fast process, but if we study it up close, we will see that overall the result is poor. It is also unfortunate to note that we are content with so little efficiency when we see what complex electronic "innovations" are applied to achieve secondary and insignificant results.

Now you understand why gas engines pollute less and have a better efficiency. Mixing with the air becomes lighter, each fuel molecule has easier access to the oxygen molecule. Burning is faster, fuller and we can really call it internal combustion.

This reason alone is enough to see that the GEET Pantone system significantly improves the petrol engine by raising it to rank of gas engine. In addition, the GEET PANTONE system allows us to drive on gas, and also looks partly on hydrogen released during the decomposition of the fuel used, which is a significant saving, allows us to drive our ordinary cars, gasoline engines, diesel, in short with public fuels without making significant modifications. It also allows you to significantly increase compression, ie. beneficial action without risk of self-ignition.

To return to the explanations on the operation of the gasoline engine, and to repeat once again, let's observe what a mixture (usually of poor quality and in inappropriate proportions) is compressed in the cylinder head is ignited by an electric spark, which causes internal combustion. This combustion inside the compressed mixture is more or less fast, more or less uniform and more or less worthy of being called internal combustion. Most importantly, if all the gasoline droplets do not find enough oxygen to burn, there will be obviously unburned petrol discharged with the exhaust gases in the form of petrol vapor or in the form of carbon monoxide (CO) produced as a result of incomplete fuel reactions means carbon from heated but unburned petrol.

In conclusion, we will say that for gasoline engines spraying is not the best solution and to apply it, we must pay close attention to the preparation of the fuel / air mixture before it enters the engine.

DIESEL ENGINE (internal combustion engine)

In this type of engine things are a little different, although it is also powered by a carburetor with fuel spray, mostly with direct injection using a high pressure pump and injection.

In essence, the engine does not accept a mixture of fuel and air, but an amount of air equal to the volume of the internal volume of the cylinder. Only after the compression of this air the fuel is injected pulverized by means of an injection (fuel dispenser, nozzle). According to the desired power, more or less diesel fuel is injected, which requires more or less time, depending on the quantity.

Entering the mass of compressed air heated by the compression itself, the fuel droplets ignite spontaneously and burn. This is burning relatively slow and lasts throughout the spray time. Fuel droplets are located in an oxygen-rich environment and react easily with him. In addition, combustion is progressive, slow and the result is better than with a gasoline engine. Of course, you have to be careful not to inject more fuel than air, so this type of engine uses compressors.

However, with this type of engine, great importance is also given to the quality of the spraying that should be as finely as possible.

So with diesel engines there is no need to pay attention to the proportions and quality of the mixture outside it, as it is able to absorb up to a certain amount of air and the fuel is injected into it after entering and compressing the air, be careful only the amount of fuel does not exceed a certain limit.

Although these explanations were not made by an engineer, I hope they helped you save time.

Successful work,

M. David

GEET PANTONE REACTOR CONSTRUCTION

Many of those who experiment with the Geet-Panton system have difficulty in implementing it, and especially in assembly of the reactor itself. They stumble upon relatively simple problems. In order not to drown in a drop of water, I give some explanations and details of the construction of the reactor.

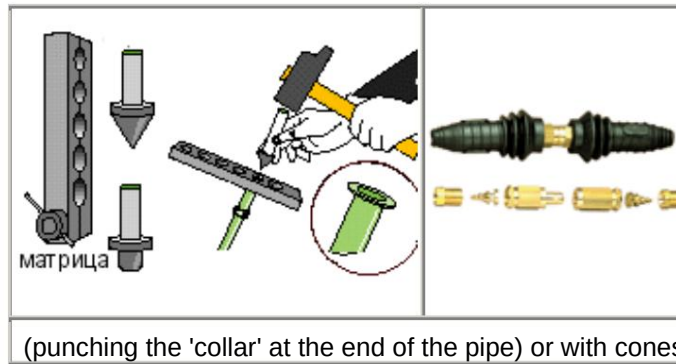
The reactor plans given on the website www.quanthomme.com, as well as those posted on the [JLNaudin website](#), have been developed in such a way that everyone should encounter as few problems as possible in obtaining the various details. However, many fail to find some details, especially a steel rod with a diameter of half an inch (about 13 mm). It is good to know that on your first try, follow all the details it is not absolutely necessary, just as you do not have to follow the way of assembly.

So, I offer you an alternative way to assemble your reactor. It is important to find an iron pipe in which it can be inserted steel rod with such a diameter that the distance between it and the inner walls of the pipe does not exceed 2 mm.; if you take 1 mm. as the maximum allowable distance, this is rather good. Of course, the length of the rod is important; if you have a lot of space, the stick it will be longer - about 35 cm, if your space is small, and 20 cm can do the job. For information I will say that my last conversion for engine of 144 cc. has a tube with an inner diameter of 10 mm, a rod with a diameter of 8 mm at a length of 16 cm. Start with the acquisition of the appropriate rod and tube, then select the other parts. For example, the iron pipe in which it is located reactor tube must have an inner diameter large enough not to interfere with the exhaust gases; if the engine has a large displacement, this must also be taken into account.

ASSEMBLY OF PARTS

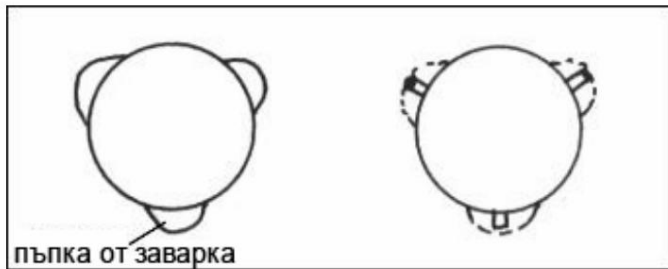
All parts can be joined by soldering, electric welding or torch welding. Avoid soldering with silver solder or with Brox (commercialized solder containing copper, zinc, tin and silicon) on all parts through which they pass (are in contact (c) the exhaust gases, as there is a risk that these parts will not withstand the heat and vibration of the engine. All welding of the parts through which the exhaust gases pass must be as airtight as possible, of the reactor to work.

The connecting nipples at the ends of the reactor tube must be welded with silver solder. Nipples can be brass with bent at the ends.



such as those used in plumbing installations. If the curved edges are of good quality, it is not necessary to anticipate sealing washers. Before welding the nipples from the reactor inlet, make sure that you can insert the steel rod through it. reactor (see diagram).

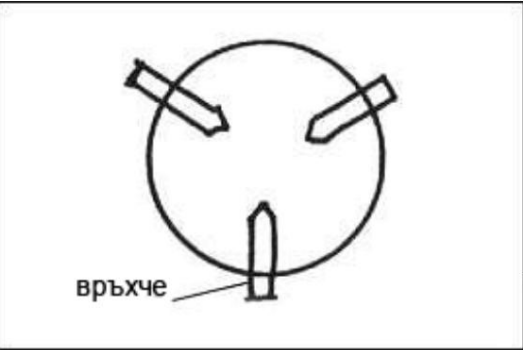
The pipes between the reactor inlet and the evaporator (tank) do not have to be metal, a rubber pipe would do the job, as the temperature of the vapor leaving the evaporator is not very high. On the contrary, the pipes from the evaporator to the engine must be metal due to the high temperature of the gases GEET (in my opinion the author was wrong - it should be read "from the reactor to the engine"). prev.) and in this case the copper fits perfectly.





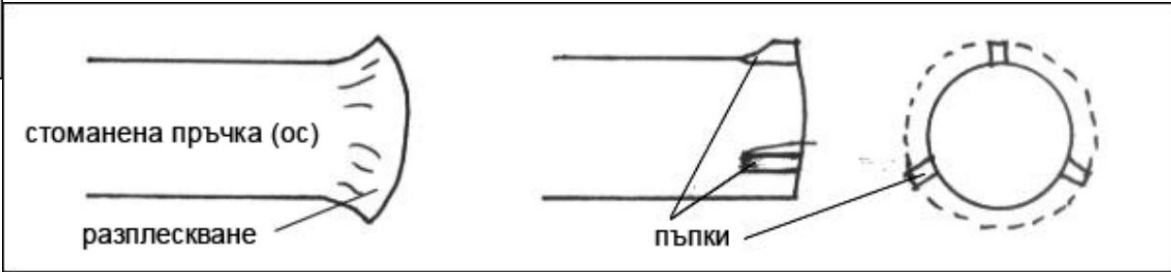
On the centering of the reactor rod (axis)

Any means of centering the rod in the reactor tube are permissible, it depends only on your imagination and the tools with which you have.

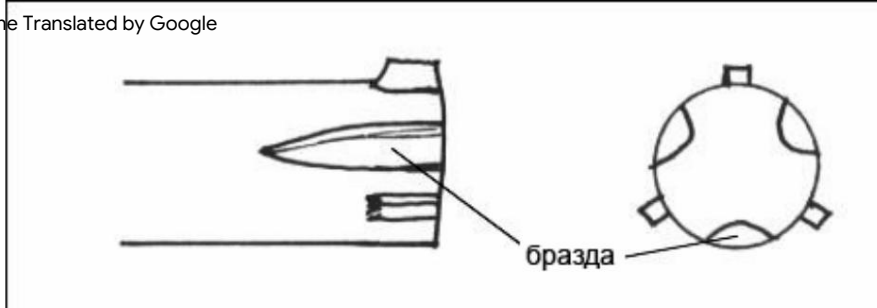


you can heat up, splash

You can drill three holes at 120 degrees and insert in them sawn metal tips with suitable length (see diagram on the right). You can also make tips (buds) with electric and drink them in the best possible way (see diagram on the left)



Furthermore the ends of



stick and drink them so that you get the desired pimples (see figure in right)

You can use other means, such as the axis (rod) to be well centered and the pimples are not worn out by motor vibrations, and can easy to insert or remove.

While you are still with the file in hand, cut furrows between every two buds to provide space for the passage of gases reduced by pimples (see diagram on the right).

Regarding the magnetization of the axis (steel rod) of the reactor

There are people who attach too much importance to this issue and complain about the weak magnetization of the reactor shaft (steel rod) rod even after prolonged operation of the reactor.

It is important to know that all iron-containing materials, except hardened steel and magnet materials, even after have been exposed to a strong magnetic field, are magnetized too weakly. Therefore, rods for the mild steel reactor (called "iron") or even STUB (hard hardened steel) will have a weak residual magnetization that can be checked with compass. If you use very low carbon steel, such as that used for your yard door, the residual magnetization will be at the measurement threshold, although the reactor will have subjected it to a strong magnetic field during its operation. This will not interfere with the operation of the reactor.

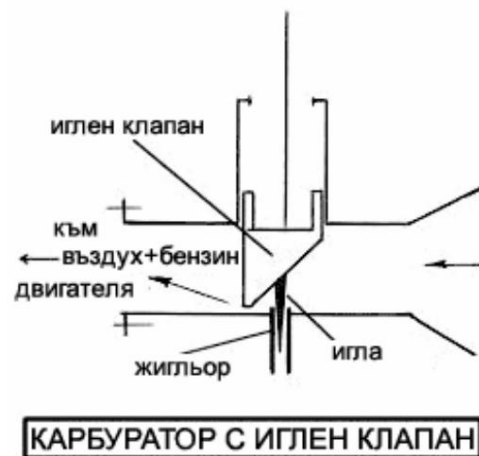
At least for the time being, the issue of magnetization should not be given too much importance. In my opinion, not the magnetic field of the reactor rod is what causes the process in the reactor, on the contrary, the latter causes the magnetization of the rod. Here the work is the same as with the color of the rod after prolonged work. However, these things can help us understand what takes place in the reactor. I will set out my point of view on this issue in a moment.

GEET PANTONE VARIANT WITH CARBURETOR INSTEAD OF EVAPORATOR

According to Mr PANTONE's patent, the reactor can be powered by a carburetor instead of an evaporator. Naturally, for this purpose the carburetor it must be able to spray petrol on as fine droplets as possible. In this case, it is not used for dosing fuel and gasoline mixture, because the dosing takes place at the outlet of the reactor, but must be able to spray gasoline finely, with a minimum amount of air. By for this reason a small carburetor with a variable nozzle should be used. This type of carburetors, called needle valve carburetors, are long used in small engines.

For those who do not know this type of hood

This type of carburetor is quite interesting and is suitable for our case, as it is a type of carburetor with varying nozzle and with varying jigger. In fact, it is not the throttle valve that controls the entry of air, but the needle valve. The difference is that the needle valve is located just above the nozzle and there is a conical needle that enters the nozzle. In this way the nozzle increases the inflow depending on lifting the needle valve and leaking more air. This results in a good petrol spray at all needle positions valve. That's why this type of carburetor is suitable for our experiments with GEET PANTONE.

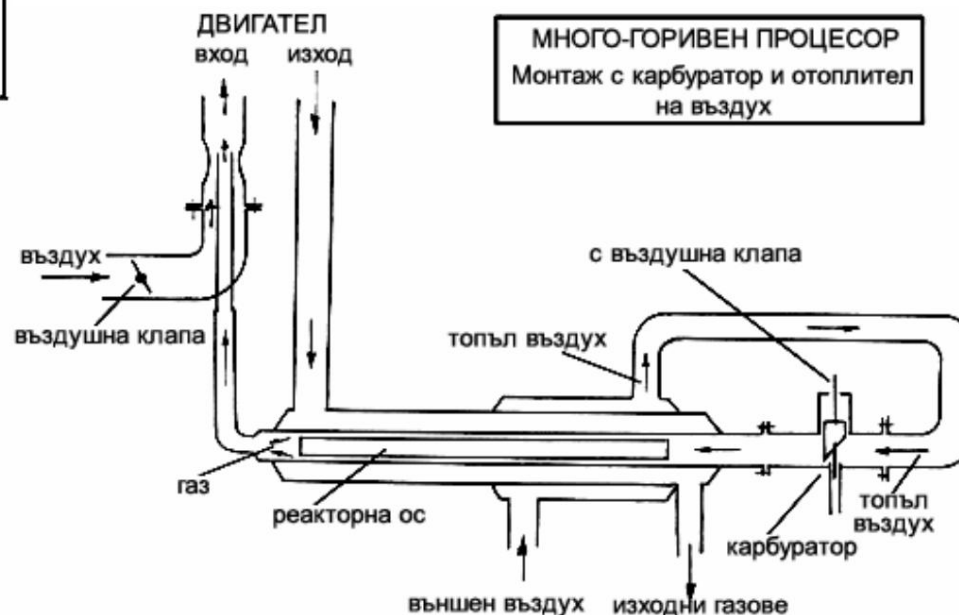
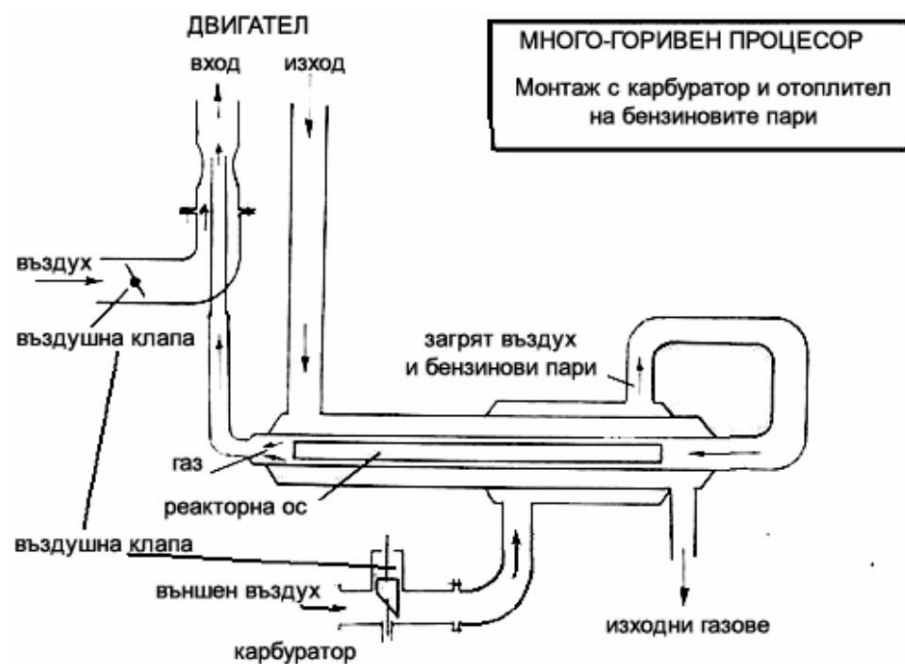


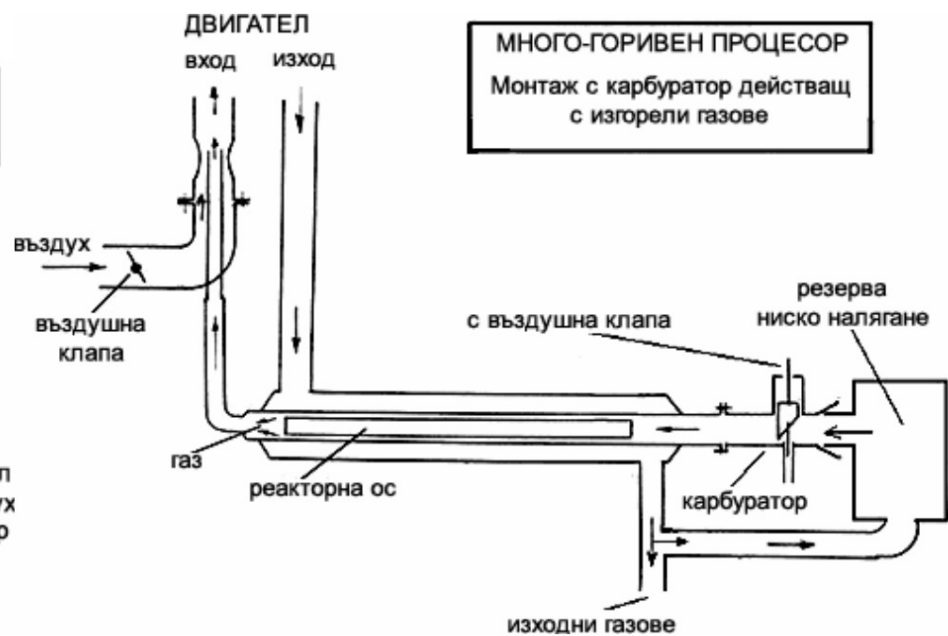
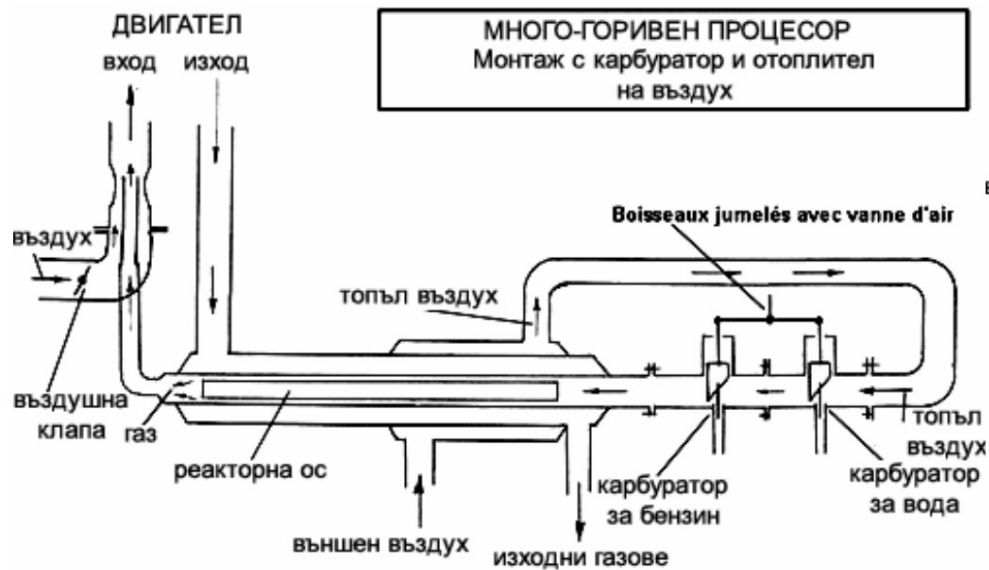
In classic carburetors, the area where the nozzle is located is a narrow part of the carburetor and is called a nozzle. It, like the juggler, is fixed, but we will find another way to deal with this shortcoming in the technology of this type of carburetors.

Carburetors with large nozzle diameters do not work to adapt to small PANTONE reactors that we want to power without evaporator. It should be borne in mind that the maximum size of the air vent in our carburetor should not exceed the opening for the passage of gases in the reactor, ie. 50-60 sq.mm for the described reactors. We can't have such a small hole for the air flow to find even in carburetors with a needle valve, but for these carburetors this is not an obstacle - it is enough to limit as much as possible possible hole with stop, stop. In any case, when adjusting your carburetor needle valve will be fixed at the optimal position according to the engine used. If possible, choose an engine with a larger capacity, at least 240 cc, to have a larger one depression in the carburetor at low operating mode, so it will be easier for you to make accurate adjustments. In any case, you will have to increase the thickness of the nozzle, as the petrol must be sprayed on the engine with very little air.

EXAMPLE OF MOUNTING A CARBURETOR INSTEAD OF AN EVAPORATOR

In the installation shown in the diagrams below, use a modified (page 2) carburetor with a throttle valve, as I explained on page 2 for the mixture from GEET gas and air and to ensure control in the various operating modes of the engine. You will install a small one a carburetor with a needle valve at the reactor inlet instead of an evaporator, and this will ensure that the fuel is sprayed with as little air as possible. Adjust the needle valve opening; this adjustment may remain permanent. To try to challenge a little evaporation of the sprayed petrol, you can pass what comes out of the carburetor (sprayed petrol) through the heating chamber installed near the reactor before being sucked into the reactor. (see diagram below). It would also be interesting to heats the air entering the carburetor in the same way. It is possible to use some of the exhaust gases instead of outside air (see scheme below).





INSTALLING A GEET PANTONE REACTOR TO AN ENGINE

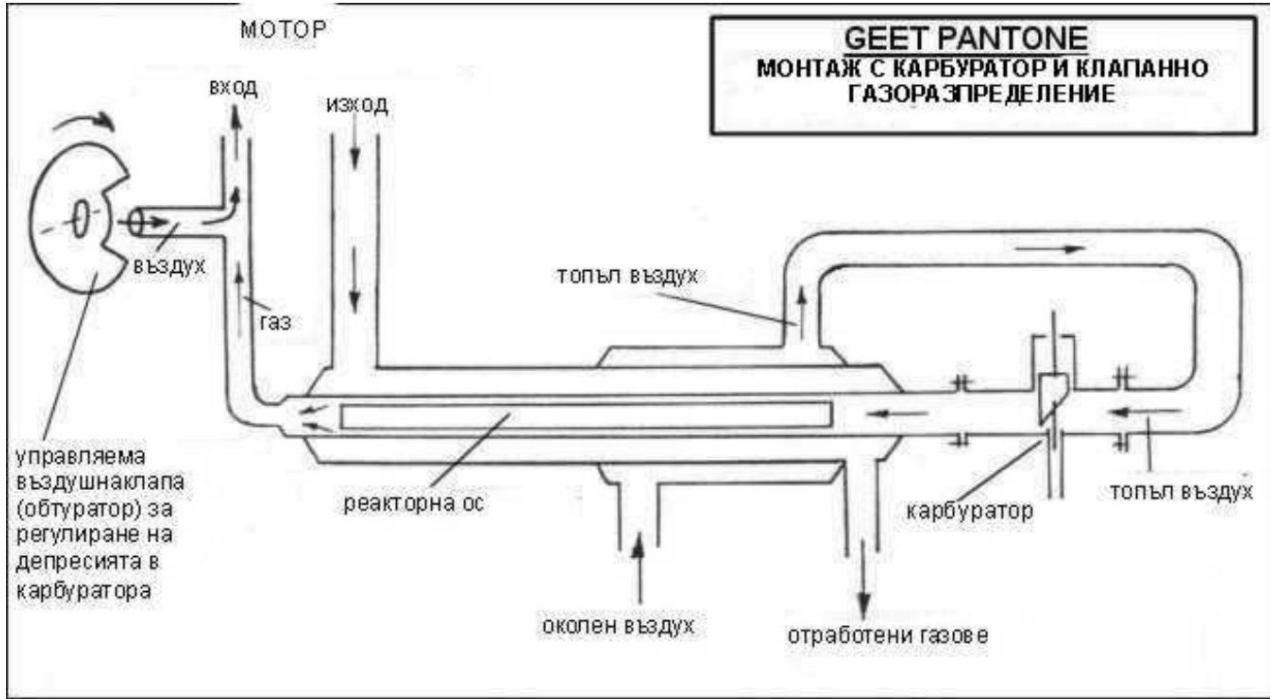
A PROBLEM YOU MAY MEET

When we want to install a GEET Pantone reactor on an engine, using a carburetor or evaporator with evaporation caused by ambient air, not exhaust gas, we will immediately notice that it is difficult to maintain it in operation at low operating conditions. The spray is difficult to maintain at a satisfactory level in the small carburetor, as the depression is weak in its nozzle, as well as in the evaporator - the depression in the evaporator is insufficient to overcome the pressure (height?) of the fuel. This problem is due to the inlet-air valve leads to poor operation of the installation and if we want more serious results, we must do the necessary adjustment.

The solution is to suck the reactor gas first and then the air one after the other, instead of sucking in at the same time, as in previous installations. Thus, as long as the gas is sucked through the inside of the reactor, the depression will be sufficient in the carburetor or in the evaporator in any engine operating mode.

In other words, while the piston descends into the cylinder in the suction stroke, the air valve must be closed at the beginning of the suction and only GEET gas to be sucked in. The air valve then opens to allow the air needed to form the detonation mixture until the piston comes down to the end.

You will see later that you will encounter problems with the mechanics, as you will have to fabricate a mechanical rotating shutter (valve) and mount it to the motor.



1- From the top dead center at position 2, the suction takes place through the reactor



2- Here only the air is sucked in, the reactor is not depressed (dilution)



3- Lower dead center before new start of cycles 1 and 2



*4- Gas / air mixing valve for
speed control
engine*



240 cm3 diesel engine with valve timing and carburetor



Le moteur 240 cm3

However, within a design designed for real use rather than experimentation, such as a car where the engine has to run well in any mode, installing a valve timing is worth the effort, as

ratio in the detonation mixture can be achieved in any operating mode of the motor and even to it to add a double centrifugal and vacuum control system.

This option, a little complicated to implement, is beyond the capabilities of an average amateur master who does not have all the necessary tools and machines. I will not go into his detailed descriptions, the concept of which may vary depending on our imagination. For those who are more at ease with electronics than with mechanics, it is possible to make a design based on an electromagnetic valve and sensors on one of the motor's belt wheels, but this solenoid-operated air valve must still be capable of to operate at motor speed.

I devised and experimented with another way to solve the problem of the air valve between the reactor and the engine. Pure and simple me I removed and connected directly the inlet of the engine and the outlet of the reactor, which I made with a larger cross section for air passage, equal to the cross section of the nozzle of the original carburetor belonging to the engine, which I used in the described experiment. In this adaptation, which is as simple as possible, which can be done on GEET PANTONE, all the air needed for the formation of the detonation mixture passes through the reactor, does not enter directly into the engine and serves entirely for the normal operation of the carburetor, it is still one thing is that the reactor is not between it and the engine. This is a very suitable solution for small engines with cylinders of no more than 300 cm³ each, as larger ones will also need a much larger reactor and several reactor tubes will have to be connected to obtain the necessary sufficient cross section for the passage of gas, **given that the distance between the walls of the inner reactor tube and reactor shaft must not exceed 1.5 mm.**

With all these installations, we will also have to preheat the air, because if it is too much, it can interfere.
of raising the reactor temperature, which is not recommended.

So this installation is suitable for very small bikes such as lawn mowers or similar, as it is not worth launching in making a rotating air valve.

Note: all these carburetor experiments were done with petrol only, but I plan to try to add a second carburetor together with the first to try as fuel water mixed with a little soluble oil to protect some small steel parts of the hood of rust.

INSTALLATION OF GEET PANTONE REACTOR TO DIESEL ENGINE

I have already given you some details on diesel engines, which are self-igniting engines with both gas oil or fuel oil.

Many of you would like to know how to adapt a GEET PANTONE reactor to such an engine. So here's some information that it may be useful if you decide to launch yourself in this operation.

At first one may think that the lack of an ignition device with such a motor can be a problem - but no, there is no a problem, as long as we keep the engine as it is when it comes to refueling. We just need to install a second one gas boost control valve so that we can control the GEET reactor and all its parts. So with these two rulers valves, the engine can function in the same way and in the traditional way, thanks to its fuel oil injection system (when the engine starts cold), and through the valve of the GEET reactor, where the ignition of the gas will be provided by the self-ignition of the fuel oil sent to the engine by the injection pump operating in the delayed position.

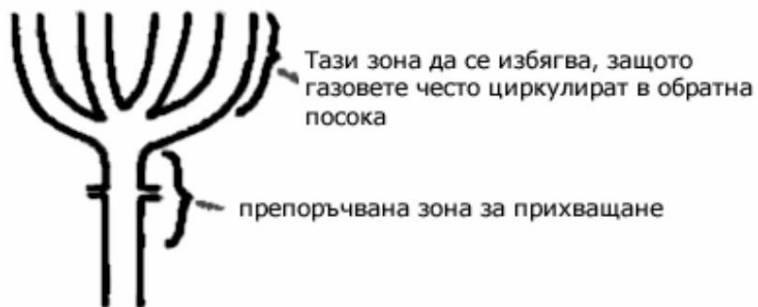
I hope you understand that the little fuel oil injected into the air / gas mixture is which after compression in the cylinder head will cause combustion of this mixture by self-ignition.

It should be well understood that while the engine is running on gas from the PANTONE reactor, it is no longer a diesel engine, but is similar to a petrol one. gas engine. When you feed it with diesel by turning the throttle lever, it works like normal diesel engine. This is possible without the risk of damaging the engine, as the reactor gas can be compressed to a large extent percentage without risk of premature self-ignition. before injecting the retarding diesel fuel which causes the ignition. Through while running on gas, the diesel engine will have an efficiency (efficiency factor) higher than that of a petrol engine with a lower compression. It is clear that the reactor must function well and not allow fuel to pass unconverted into gas.

WHAT IS THE WAY OF PROCESSING THE DIESEL ENGINE REACTOR



**ПРИХВАЩАНЕ НА ОТРАБОТЕНИТЕ ГАЗОВЕ ОТ ИЗПУСКАТЕЛНАТА
ТРЪБА ПРИ МНОГОЦИЛИНДРОВИ ДВИГАТЕЛИ
(бензинови или дизелови)**



It is preferable that we have already mastered the construction and operation of the reactor and additional parts on petrol engines, building on the previous ones by notes and your own experiments on this topic.

To make your diesel engine work with the PANTONE reactor, you have two options - the use of gasoline or diesel fuel. For starters I advise you to choose gasoline, as you have undoubtedly collected more before experience with it and it is very likely that your experiments were successful. Using diesel fuel creates more problems with evaporation, so you can try it later.

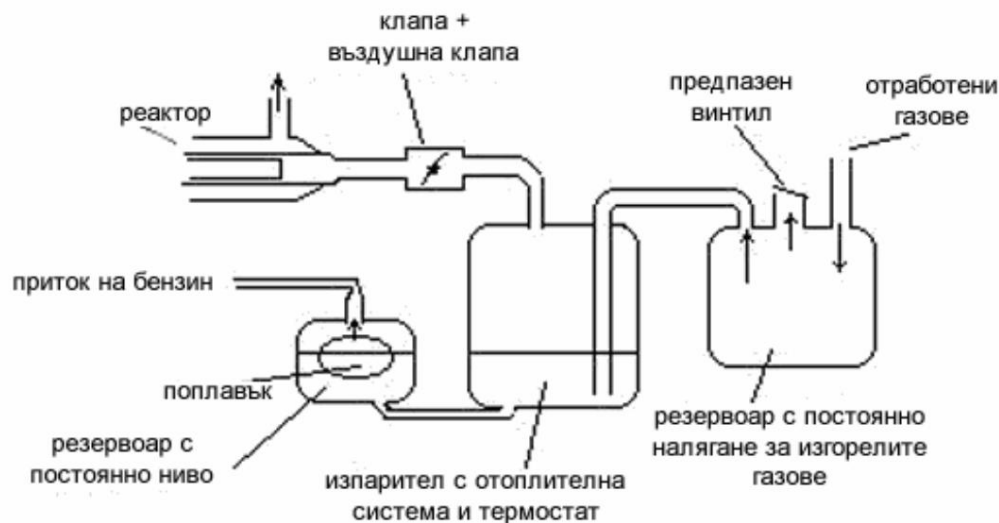
In order for your reactor to run, you need to divert some of the exhaust. You can do this without damaging the exhaust pipe too much (muffler) adding an additional part where the outputs are connected pipes. You can make this T-shaped part yourself with the same diameter soldering the corresponding sealing cones and a piece of pipe with a smaller diameter that you will connect to the reactor.

If you have designed your device with an evaporator, you will need it keep warm at a constant temperature. You have two solutions - a) heat with the help of the exhaust gases by means of a coil and a thermostat, as described in the previous pages; (b) or with water from the refrigerant of the motor you run through the heating coil. This way you can maintain or change the desired temperature in the evaporator, which is practical when experimenting with alcohol, gas oil or various oils.

At the engine inlet, the air lines from the four cylinders are collected together with the air filter, so you need to provide for this place the connection to the outlet of the reactor, which also has its own air valve, which will take air to the air filter. It should be noted that when the engine only runs on diesel, this air vent or someone

another must be fixed in the maximum open position. The installation of two air valves can be envisaged - one that will be either fully closed or fully open when the engine is running on diesel, and another that serves only to mix the reactor air and gas. These two valves can be cleverly made using a double-body carburetor. One body will be converted into a gas carburetor, and in the other body the throttle valve will be equipped with an opening device connected to the actuator diesel injection pump mechanism.

**ПРИМЕРЕН МОНТАЖ НА ИЗПАРИТЕЛ ДЕЙСТВАЩ С
ОТРАБОТЕНИ (ИЗГОРЕЛИ) ГАЗОВЕ**



If you select the evaporator system with an evaporator using exhaust, it will be appropriate to install a tank with constant pressure for these gases for the purpose of stability of the operation of the evaporator. Such a tank can be made of metal a few liter can and put it regulating valve (see diagram).

All you have to do is get to work and try to use it more imagination than me. Don't forget to get the most out of it safety measures for experimentation, for prevention of accidents.

See also page 9, where I offer you a simple solution suitable for both diesel and petrol engines.

- PAGE 7 -**THE EASIEST WAY TO INSTALL A PANTONE REACTOR IN DIESEL OR
GASOLINE ENGINE**

A farmer from my area invented and installed on three tractors this option, which is simple and effective. This approach represents an interesting compromise compared to ordinary installation due to its easy implementation. So instead of looking for a way to spend through the reactor at the same time the fuel vapor and possibly the water, we spend only the water vapor and supply the engine with normal fuel (petrol or diesel) without modifications to the installation. Thus, the inflow of fuel gas leaving the reactor, probably learned from the decomposition of water alone strengthens the engine and improves the combustion of the classic fuel that the engine runs on, be it diesel or petrol. In the case of diesel fuel, the engine operates as a petrol engine (moteur à explosion) rather than as a diesel engine (moteur à combustion) and this phenomenon occurs when fuel is injected. We find that the inflow of gas obtained from the decomposition of water brings significant additional power to the engine, the consumption of conventional fuel decreases significantly, burns more efficiently, which leads to reduce environmental pollution. The observed fuel savings vary by about 25% for two of the diesel tractors, and much more significant for the third, who was in extremely poor condition and thus found his "second youth." In all cases, after reaching normal operating temperature, the engines of these tractors do not produce any smoke or odor.

As far as I am concerned, it remains for me to clarify one point concerning the combustion of water from a motor equipped with a PANTONE reactor. The reactor Does the water molecule actually decompose into hydrogen and oxygen, or does it only produce high-temperature water vapor, which decomposes in the cylinder under the influence of the high combustion temperature of the conventional fuel. This remains to be seen.

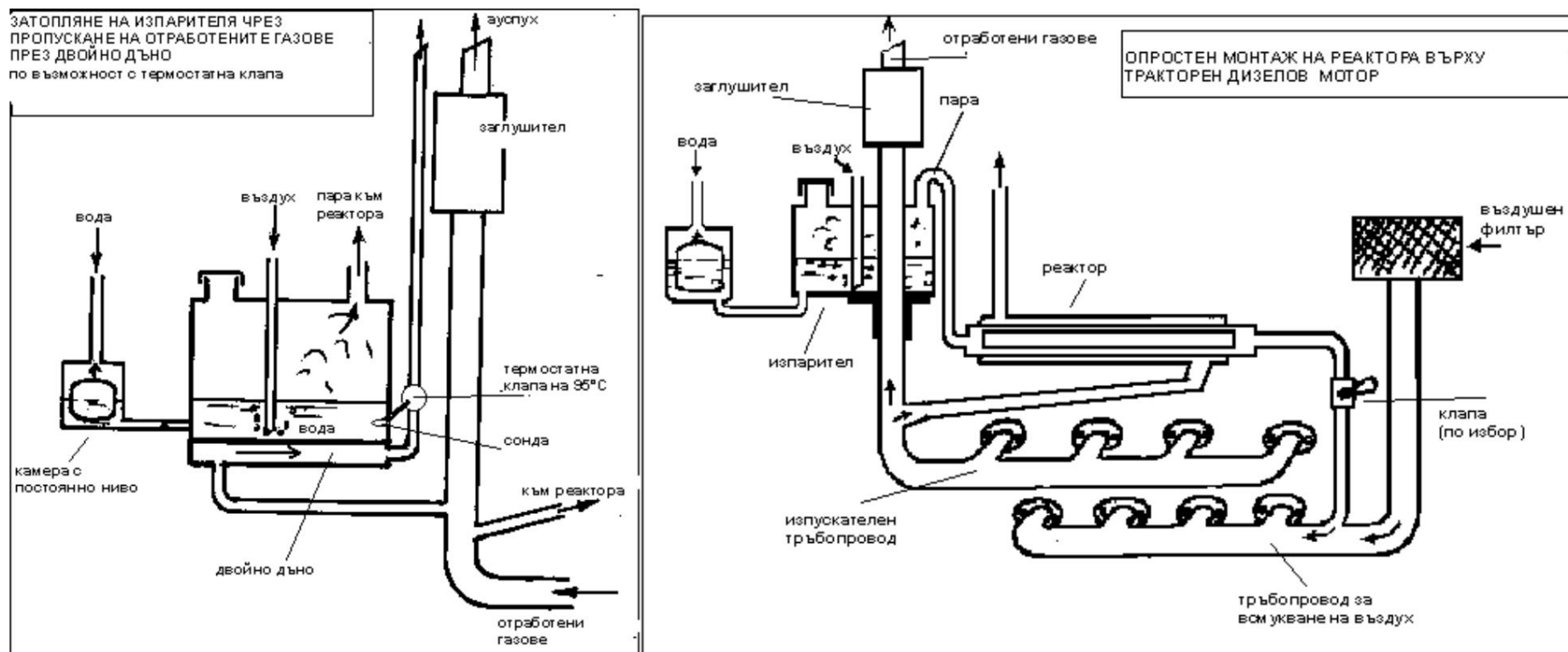
**DESCRIPTION OF THIS SIMPLIFIED INSTALLATION MADE IN
TRACTORS.**

Note: Nothing has changed in the motor at the injection pump.

- The evaporator is welded around the exhaust pipe before the muffler (see diagram and photos). • A metal tube exiting the evaporator is connected to the reactor inlet and the reactor outlet is connected to the suction line.
air.
- The reactor is installed where there is a place as close as possible to the exhaust pipe, in order to connect the pipe to be as short as possible and make maximum use of the hot exhaust gases.

In this simplified version, where only the water evaporates and its vapors are sucked into the reactor, we put only water in the evaporator. Must we manage to keep the water at maximum temperature, but without boiling, in order to evaporate more easily through the air that passes through her. A temperature with an upper limit of 90 ° is perfectly suitable. It is best to make an evaporator with a permanent thermostatic valve

temperature, as described on the second page, which regulates the flow of exhaust gases in the coil, or even easier - in one double bottom (see diagram). Another option is to use the water from the engine cooling system to keep the evaporator warm. (using a double bottom) instead of the exhaust, which has the advantage of eliminating the problem of regulation, as the temperature of this water is about 90 °. But if we use this option, the time to reach this temperature will be longer than with the exhaust gases. In this case, the connection is made as close as possible to the cylinder head before the thermostat.



A power supply must be provided to automatically maintain the water level in the evaporator using a float chamber, as at the carburetor. In a vehicle in which we cannot install a water tank higher than the evaporator, it is possible to we supply the evaporator with a small wiper pump driven by a contact float placed in a small tank connected to the evaporator (contact float, such as in the master cylinder reservoir).

The evaporator does not need to be large in capacity if it is constantly powered. A two-liter can in which we pour half a liter water is quite sufficient. Personally, I plan to try to replace the evaporator with a small carburetor, but I do not have enough time to do this experiment. If this option works, the work is much easier. Despite the ease of installation made by this friend farmer, with only water flowing through the reactor, this option must still be improved, subject to certain conditions:

- The water temperature in the evaporator must be kept constant at approx. 90 °.
- The air that causes evaporation, passing through the water in the evaporator, must be preheated, especially in winter time ..
- The reactor must have the highest possible temperature, at least on its outlet side.

For his largest tractor, this farmer installed several reactors (7) inside the jacket of the exhaust pipe in parallel.

Note: the jacket containing the reactor (s) and the exhaust pipe must be large enough not to clog exhaust gas circulation, which would slow down the engine.

I think that the number of reactors should be determined precisely so that the operating temperature of the installation does not decrease too much, especially when the engine is running in low mode. All this is possible through experimentation.

- PAGE 8 -

USING THE GEET PANTONE AS A BURNER



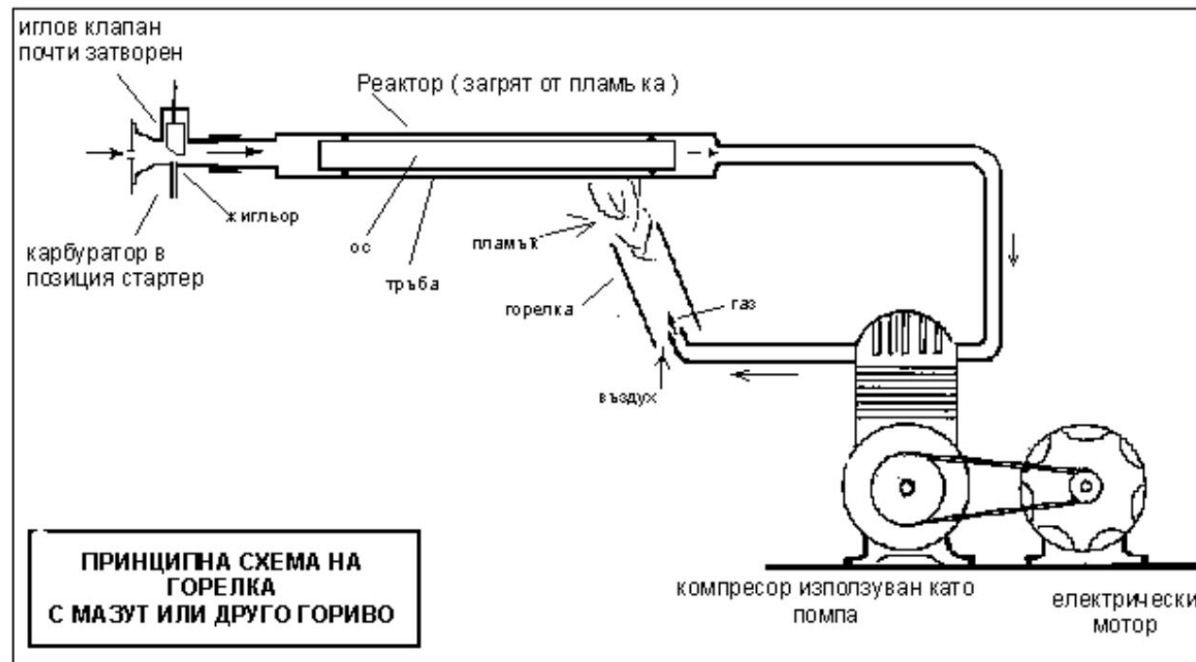
According to Mr. PANTONE's recommendations, we can use his reactor to make a burner that uses different types fuel.

First we will put together a small prototype that can help you evolve to make a boiler burner in central heating (fuel oil) operating in "either zero or maximum" mode or in continuous mode.

Attention: for safety experiments with this version of GEET must be done outdoors, the responsibility is entirely yours.

To do this experiment, you need materials that are affordable and inexpensive:

- small air compressor with motor.
- small carburetor from a 50 cm³ moped.
- PANTONE reactor made of pipe and steel shaft (see previous pages).
- several copper pipes and couplings.



Comments and functioning

After looking at the scheme, we see that the work is so simple that it does not need to be explained.

The reactor is actually a steel shaft (rod) 30 cm long and 14 mm in diameter. It is placed in a tube with an inner diameter of 16 mm and with length 40 cm.

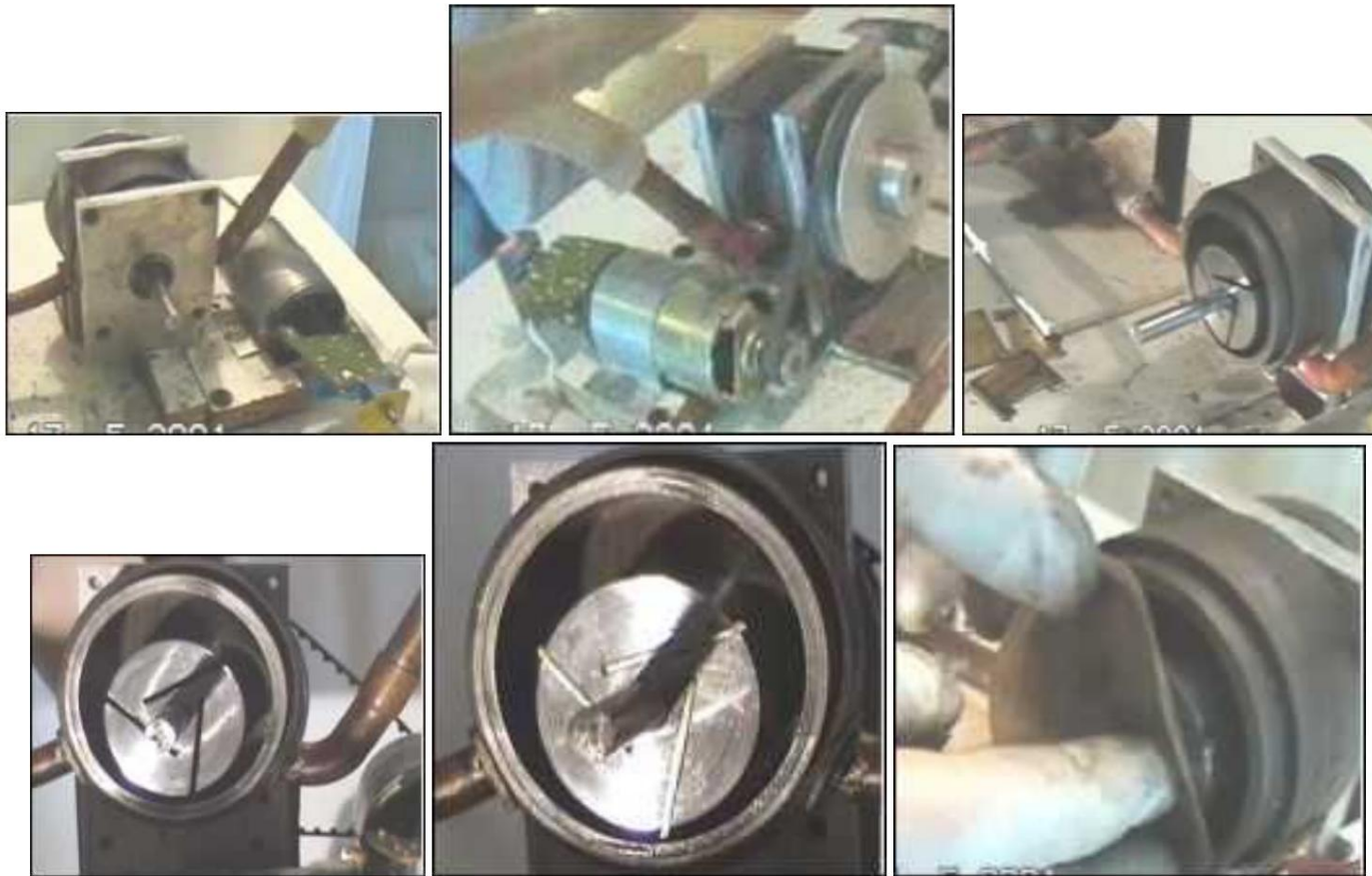


At one end is the carburetor, which will operate in the "starter" position if the compressor is small and not sucks a lot.

Pipes 12 or 14 (inches?) Connect the reactor outlet to the suction side of the compressor and the outlet of the compressor (copper pipes) goes to the burner, made of a piece of pipe with a large diameter. The concept of the burner should be adjusted to the flow rate of the compressor to keep the flame stable.

You also need to look for and adjust the carburetor adjustment, which also depends on the power of suction of your compressor.

If your suction seems too strong, you can change it with a few belt wheels. In order not to pulsate a lot of flame, it is better to take a two-cylinder compressor. As you can see in the photos, I used a turbopump that I made myself, but not everyone there is a workshop available to make such a pump.



IMPORTANT: never insert gas reserves between the reactor and the compressor and especially between the compressor and the burner to avoid any risk of explosion.

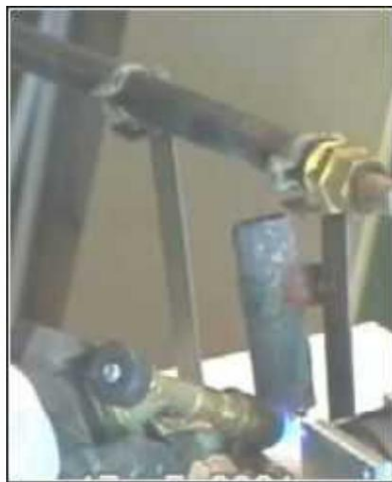
FUNCTIONING

This installation starts without a problem in the cold with fuel oil (for boilers central heating).



This installation starts without a problem in the cold with fuel oil (for boilers central heating).

When everything is OK, narrow the carburetor to the "starter" position and start the compressor.



Yellow flame at the beginning.

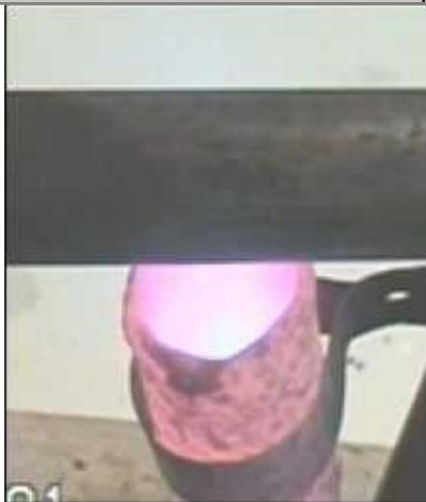


Blue flame after 1 - 3 min.



Blue flame up close.

Bring the flame closer to the base of the burner with ignited oxygen, it will ignite with a yellow flame and may continue to burn if you hit the exact proportions of the burner. If not, you will have to finish it so that the flame does not go out.



Below the reactor, the flame inside the burner. The flame temperature is quite high (about 1200 ° C) as it glows to a white steel rod.

Steel platinum and steel rod glowing to white in a few seconds

I will continue this experiment in order to use some amount of water as fuel, as with engines - either with a carburetor or with evaporator.

Next ...

TIPS AND IDEAS FOR MAKING WATER TANKS-EVAPORATORS FOR PANTONE REACTOR

For a successful installation it is necessary to pay special attention to the evaporator system.

Here are some tips and solutions to save time in making these water evaporators, which are used in all reactors PANTONE.

Let's go back in detail to the necessary conditions so that we can hope to "burn" water in all types of engines equipped with GEET PANTONE reactor.

- water vapor must enter the reactor as "dry" and hot as possible, mixed with a certain amount of air; • more precisely, the reactor must be sucked in as hot air as possible, carrying a certain amount of humidity - we achieve this by passing air through water at 90 ° C, also at 90 ° C; • Bulk water vapor or droplets should not be sent to the reactor, as this may cause the reactor to cool down. and not get the desired result.

In short, the water in the evaporator should be as hot as possible, but not boiling. The air should also be as hot as possible, but below 95 ° C, not cause the water in the evaporator to boil and flood the reactor.

The water in the evaporator must be kept at a constant temperature.

The air must also be at a constant temperature.

A device must be provided for the evaporation to take effect only when the water in the evaporator has reached its engine starting temperature. This device is also necessary to avoid the ingress of untreated water through the reactor in the motor. Reaching the required installation temperature takes only a few minutes, during which time some may be puzzled. from the appearance of white smoke (steam) coming out of the muffler.

Although we find interesting results without adhering absolutely to these tips, it is best to fix as much as possible. may more parameters to be functioning "top" in all conditions. This allows various serious analyzes to be performed.

To heat the evaporator you have a choice between two options:

- heating by water from the motor
- exhaust gas heating

Heating with water from the motor is very easy to do as it is enough to drain the water at the pump outlet before the thermostat. At a car just needs to connect to the heating pipes system with copper T-shaped. Water maintained at a temperature of about 90°C thanks to the thermostat of the heating system, it will heat the water in the evaporator by means of the double bottom, which must be large enough due to the heat it must give to the installation (see diagram on the right).

Raising the required temperature is a bit slow in this installation, but still construction is easy. The evaporator tank must be made of stainless metal (copper, zinc, bronze, etc.) no leaks. It would be appropriate the evaporator is thermally insulated in some way.

Heating the evaporator with the help of exhaust gases is more complicated for making and its only advantage is that the temperature rises faster.

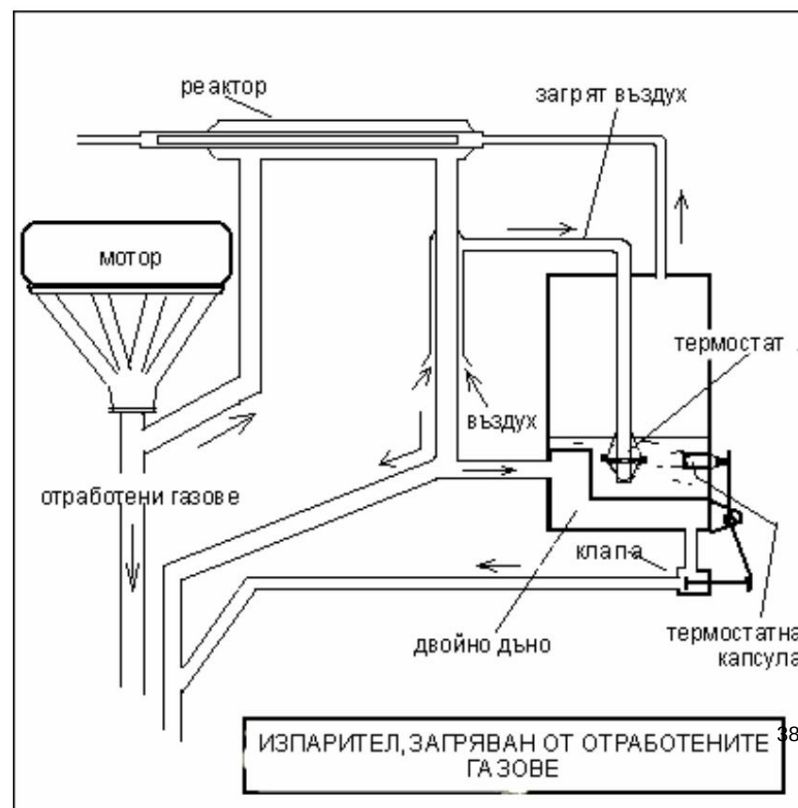
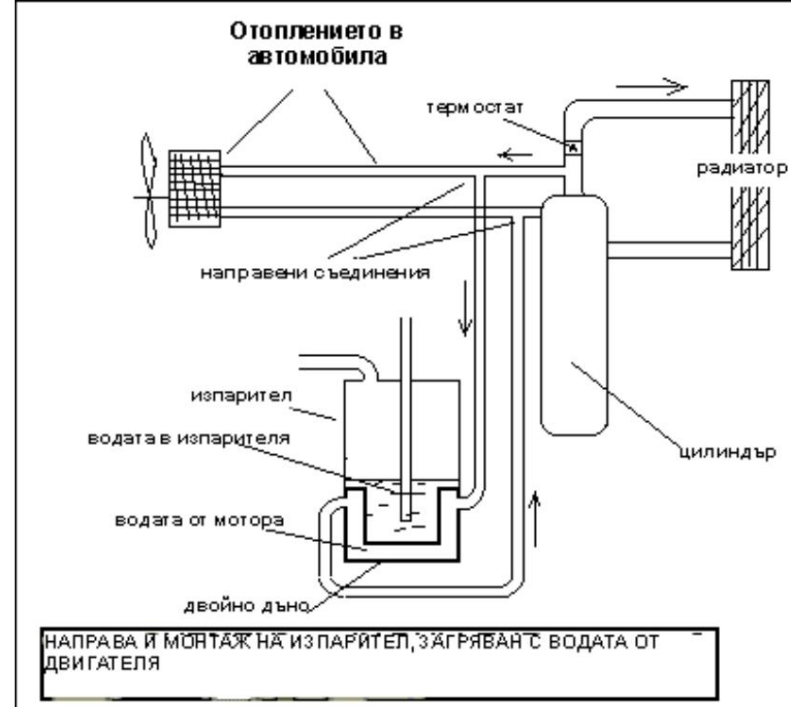
The evaporator is made on the same principle as the previous one, but the double bottom will be less, of metal and with solder, resistant to exhaust gases.

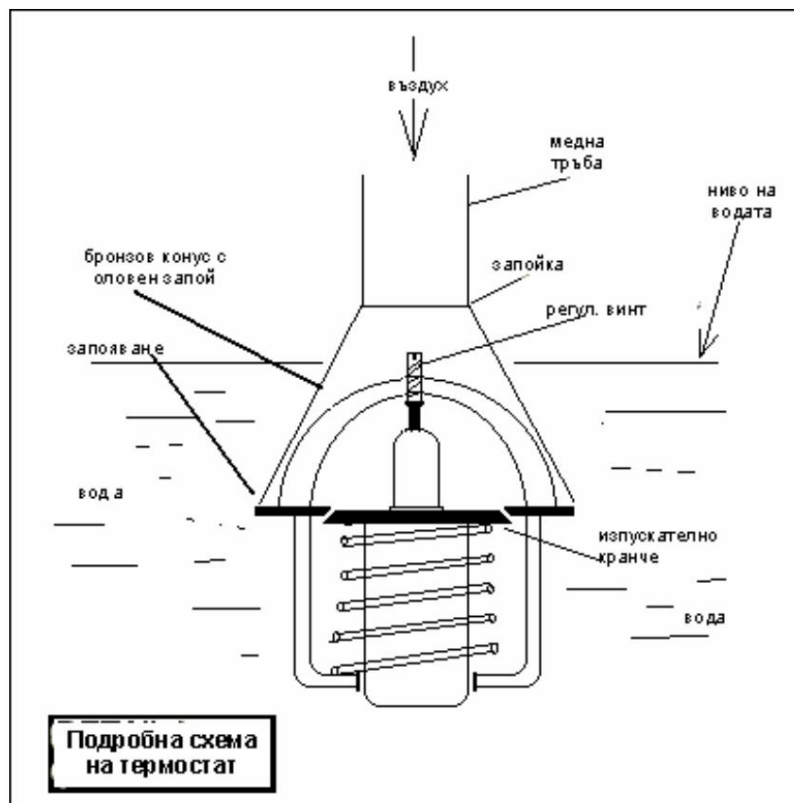
The main inconvenience, along with the numerous metal pipes for making that the temperature of this device must be adjusted one at a time system, which I have already described on the second page, to avoid rapid boiling of water.

As you can see, this type of evaporator system is quite complicated to make and not you I highly recommend it, but here's the schematic.

The issue of preheating the air intended to pass through the evaporator must be resolved.

Heating this air is mandatory, but at the same time it is a parameter that is easy to fix. The operation of the evaporator will be better, especially in cold weather.





opening at 88 ° C, most often placed in the reinforced rubber tube of the output of the water pump.

You need to modify the opening threshold of this valve by tightening the adjusting screw on the capsule. This is done in a pot of hot water and with thermometer. Adjust so that the valve starts to open at water temperature about 80 ° C. Thus, evaporation will begin slowly before the ode in the evaporator to reach 88 ° C.

Since this thermostatic valve is made of bronze, you can easily solder it with lead on a funnel of bronze or red copper, and the funnel itself soldered at the end of the evaporator tube, which is also made of copper (see diagram on the right). If you are chose the system of heating the air with the water of the evaporator, do something like a "bundle" of pipes that will be located in the water of the evaporator

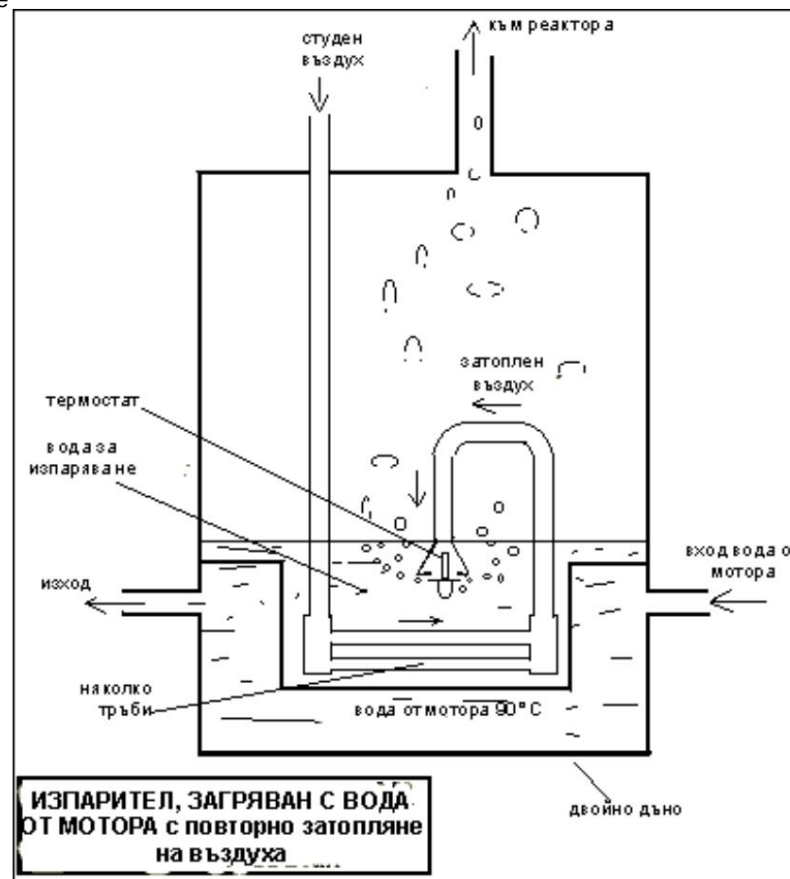
You can heat the air by passing it through the exhaust pipe, but the temperature will be variable. You can also heat it in or in the water of the evaporator, or in the water in the double bottom by means of a "bundle" of pipes (see diagram).

It remains now to make the device that allows evaporation from the moment the water reaches the desired temperature.

As I told you, you should avoid sending to the reactor Insufficiently heated money, so I recommend that you use a device that opens the flow of money only when the water is almost the same engine temperature.

You can make this device very easy by customizing one thermostat from the engine cooling system at the end of the tube immersed in the water of the evaporator.

Here is the scheme of one such a thermostat that is nothing more than one exhaust faucet (valve),



or in the double bottom, through which the air will pass, and this "bundle" will be connected to the evaporator tube (see diagram on the left).

Be sure to provide power to the evaporator as described on [page seven](#). 

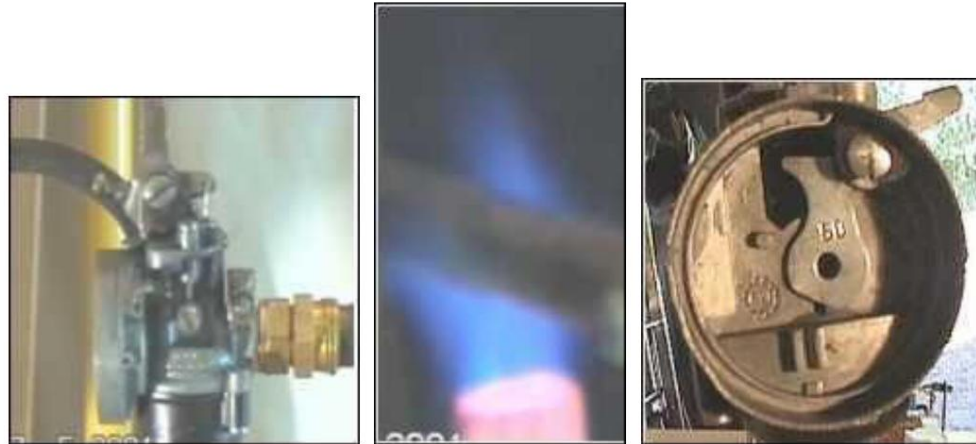
Of course, all these instructions for making an evaporator are just a few personal ideas, I count on you to find others and better.

Although the evaporation of water gives great results, I plan to try other means based on carburetors to obtain the same results, it takes up less space under the hood and may be easier to do.

- PAGE 10 -

FUNCTIONING OF INTERNAL COMBUSTION ENGINE WITH DIESEL OR FUEL WITH PANTONE REACTOR ASSISTANCE

You've already seen on [page eight](#) that my experimental burner works perfectly with fuel oil and makes a nice blue flame like gas burners, with no smoke or odor from the moment the reactor is heated with fuel oil sprayed with a small gasoline carburetor.



As in the above experiment, I acted in the simplest way - because this small one carburetor is not heated, but on the contrary, even cooled by the spraying of fuel oil, it must put in the "starter" position to maintain this spray.

This is how my small 240 cm3 bike, which serves me for all experiments, works perfect with this small carburetor and with pure fuel oil (for heating).

Because it is not possible to start cold without the reactor reaching a certain level temperature, we must start the engine with gasoline to warm up for 2-3 minutes, and yes switch to fuel oil after that.

We find that the engine works as well as with gasoline and does not produce either smoke or odor from the exhaust pipe.

In this case, the engine works exactly with the gas produced during the decomposition of fuel oil in the reactor, as we see with the naked eye by the color of the flame. on the burner - the content of carbon monoxide (CO) is zero and there is very little carbon dioxide (CO₂).

Although you are not legally allowed to drive your petrol car with gas oil or fuel oil transformed into gas through the PANTONE reactor, it is still interesting to use the economical operation of the motor to power the electric generator, small tractors and other machines that they usually consume a huge amount of fuel and are ranked as the biggest air pollutants.

To adjust the engine of your choice, you must consider its cylinders to determine the dimensions of your reactor.

Always design the reactor as small as possible so that it can heat up faster and maintain the temperature even in slow motion. motor stroke. Make good thermal insulation at the exit of the exhaust gases and at the exit of the reactor with glass wool or other high temperature resistant fabric.

You can keep the original carburetor of the engine, which will work maybe well without changing to the "starter" position, but in all likelihood, in order for everything to be all right, you will have to modify the float adjustment so that the level of fuel oil in the float camera as high as possible, naturally, without overflowing through the nozzle. You may need to put a smaller or larger nozzle, that you will determine by guessing the stages. You also need to guess the position of the needle valve or throttle valve, the quantity incoming air, etc.

If you manage to make the carburetor heat up slightly, as well as the air passing through it, it will be best.

As you can see in the photos, when my engine was running with a carburetor placed at the reactor inlet, I installed my rotating air valve a valve that mixes the incoming air with the gas leaving the reactor, the necessity of which I have already explained [on page five.](#)



rotating air valve

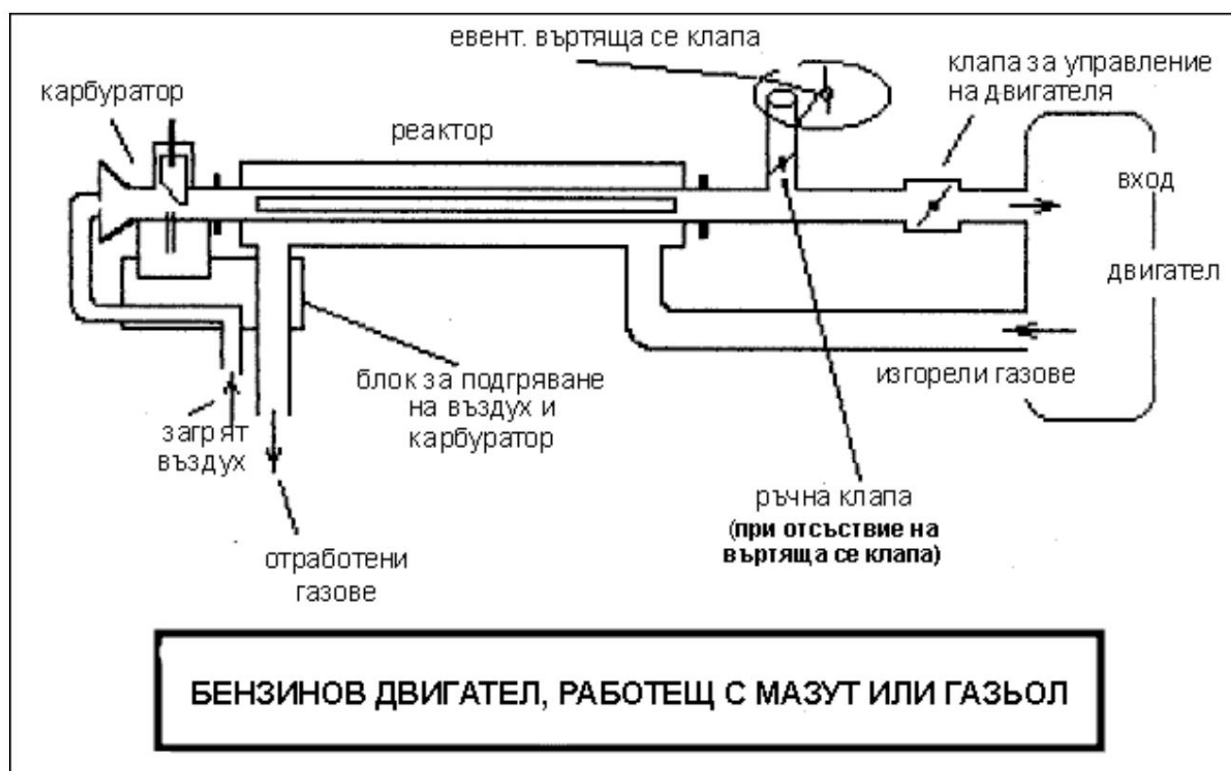
It should be understood that by temporarily blocking the flow of fresh air, we increase the suction through the reactor, and therefore in the carburetor, with in order to have the necessary fuel spraying even at low engine mode.

If you work with a motor on which the normal operating mode is relatively high, such as an electric generator, you can go without rotating air valve.

Of course, you can think of another way to increase the suction in the reactor and the carburetor, respectively. This is a problem that it is difficult to solve in a simple and efficient way, and I am still looking for a better solution to offer you.

As I told you in the first pages, [making](#) the [PANTONE](#) reactor itself is not a problem, but the peripheral equipment is one that is difficult to adjust due to the many options available.

Now, if you have the necessary competencies, you can replace the "carburetor" solution, accessible to all, with the "injection" solution - with pump, electronic pulse generator, sensors, etc. This solution I chose for my boiler on the central heating boiler, which will I will describe you in the next page.



WHAT YOU NEED TO KNOW ABOUT THE FUNCTIONING OF CARBURETORS AND THEIR SUCCESSFUL ADAPTATION TO PANTONE REACTORS

It has been several months since I wrote the first ten pages about the GEET-PANTONE reactor. I warned at the beginning that the design of the patented reactor is not a problem, but that all the difficulties in adapting it to a given motor consist in the concept of additional parts and devices necessary for its operation.

If you read the previous pages carefully again, you will find that I have listed these difficulties and suggested some solutions for overcoming them.

As I told you, the evaporator system is not the best solution for equipping a reactor designed for a car or a permanent one. installation, although it is easy to do in order to convince the efficiency of the invention.

After the first experiments, it is better to focus on the "spray" system with carburetor or injection.

I will soon make a page dedicated to injection mounting, which is the ideal solution, but you must be able to handle electronics, to be able to install a pump, etc. After all, we don't need to use this sophisticated technology for little ones single-cylinder engines, especially when operating at a constant speed (eg electric generator).

So it is good to keep working on carburetors, provided we are fully aware of what we are doing and what the results are.

we are looking for.

Adapting a normal carburetor to the PANTONE reactor is a delicate job for the simple reason that the vacuum (depression, discharge) and flow in it are very weak due to the air flow just before the inlet to the engine (suction).

I have already mentioned this important problem and described a system with a rotating air valve (shutter), which closes for a moment the flow of air during intake to obtain maximum depression (vacuum, discharge) and the carburetor to function.

This device is satisfactory, but you must have a workshop available, be able to design such a mechanism and install it to the drive shaft.

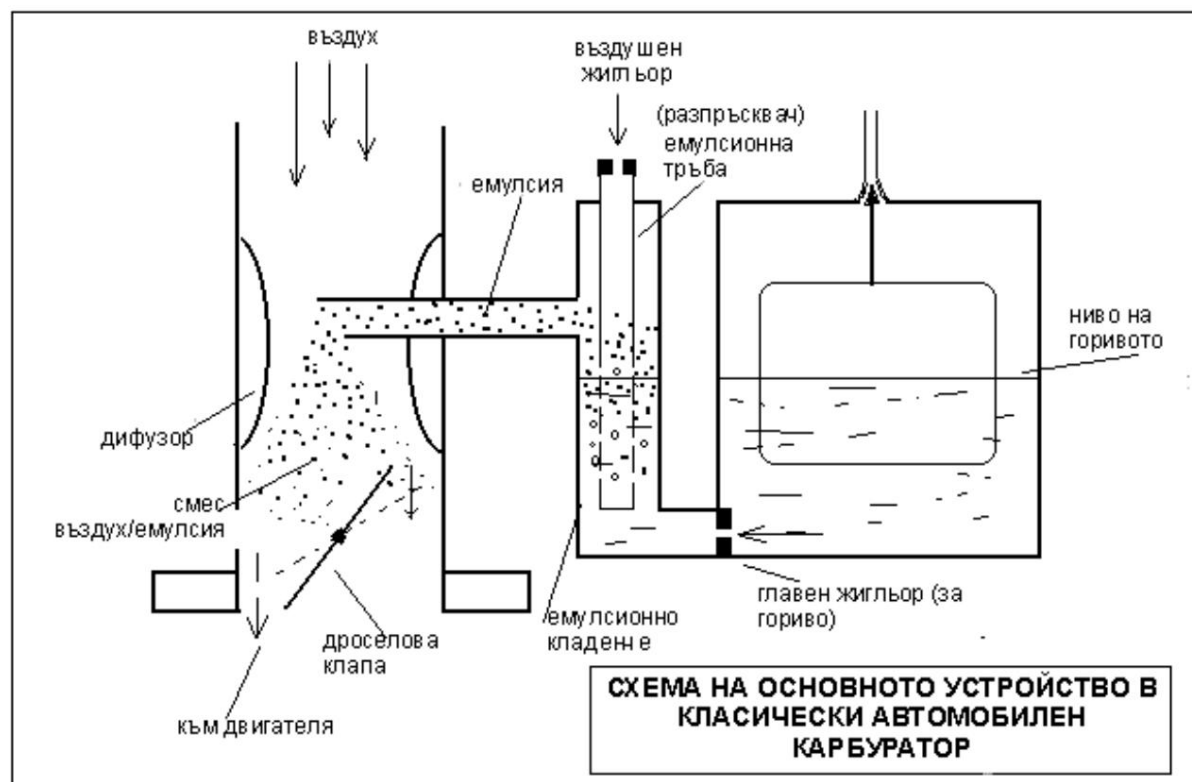
For these reasons, I think it is necessary to describe to you now a way to get a good result without a rotating air valve (obturator) based on a classic carburetor.

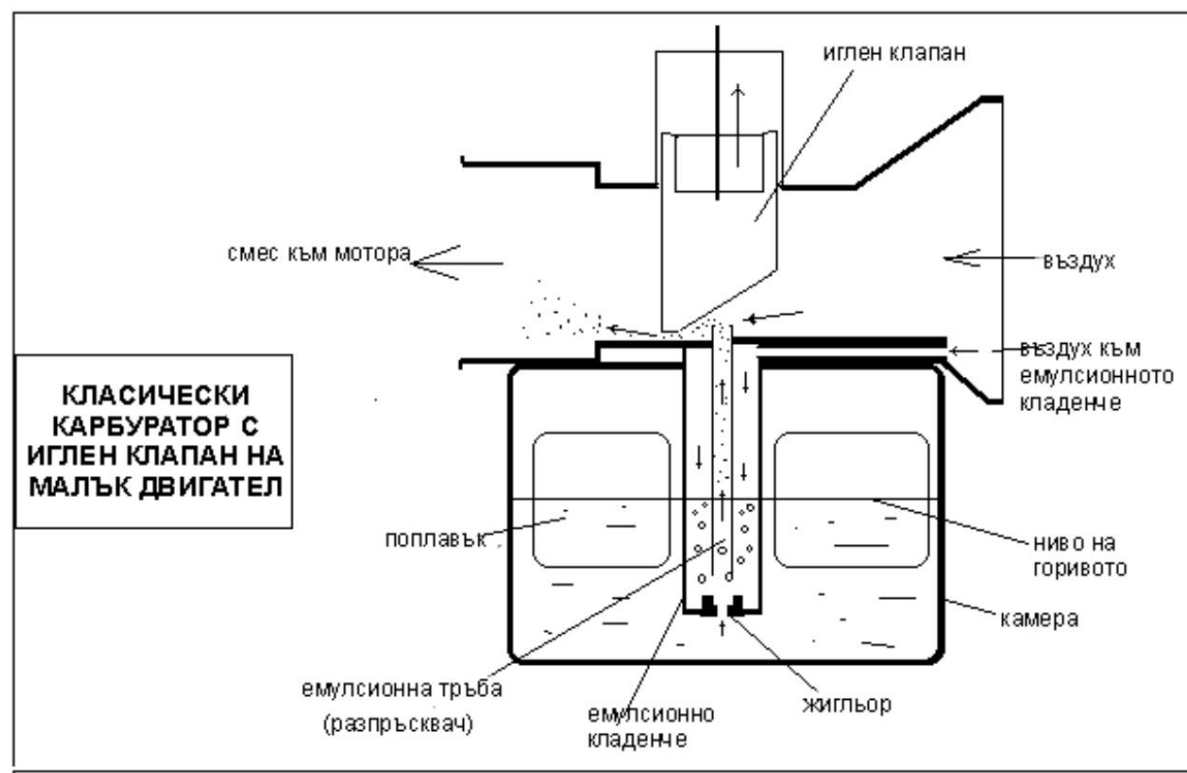
Note, among other things, that the real concepts in new engines equipped with PANTONE reactor can be imagined with two intake valves - one for GEET gas injection only and the other for air supply, with some difference in opening time. On this one way we will have severe depression in the reactor, the same thing I achieve with my rotating air valve.

We can also imagine a free rotating shutter, not connected to the drive shaft, which makes things a little easier. He is made like a turbine or a fin on which the blades clog for a moment, rotating themselves by air circulation. Even though he is easily, this device must be made carefully, with small bearings, so as not to waste energy and be able to work at high speed. Personal i made a shutter with the motor and and the plane of a hard drive from a computer that has a rotation of very good quality.

Let's start with a brief reminder of the operation of a carburetor, which in fact few people know in depth. I will leave aside additional devices, such as the idle device, for the initial start-up, to increase the speed and others that are not interesting to us to focus on the basic device, say the emulsion well:

EMULSION WELL - principle of operation





During the suction stroke, the discharge (depression) makes sure that the outside air is sucked in through the diffuser, the discharge (depression) also absorbs the emulsion that is produced in the emulsion well, the extraction of the emulsion is enhanced by the effect venturi produced by the air circulating in the diffuser. If we look more closely, we find that the emulsion well, with its emulsion tube and its two nozzles representing a real small evaporator.

Since this evaporation takes place with very little air, we have an aerosol rather than vapor coming out of the emulsion well.

This aerosol, consisting of air and fuel, receives in the diffuser additional air necessary for complete combustion in the engine.

The emulsion well with its two nozzles is the most important part of a carburetor, as it is in itself a real an automation device necessary for the production of the necessary air-fuel proportions in any mode of operation of the engine. In fact, if we remove the emulsion tube with its air nozzle, keeping only the fuel nozzle, we will get carburetor of the paint gun type. With such a device and a very small nozzle it will be possible to move on satisfactory engine in a certain mode, but when accelerating the air-fuel mixture will lose its balance (proportions si) and too quickly you will get too much fuel in the mixture, which is not our goal. Such a carburetor does not fit a variable engine operating mode.

So obviously with such a device, the faster we accelerate the engine, the faster and more air is sucked into the nozzle, while drawing fuel in increasing proportions, and what we need is that these proportions are constant in any engine operating mode.

Leaving aside the extremely fine spraying, here we are facing the main problem with the carburetor, as it is necessary to send to the engine in any mode of operation an accurate and constant air-fuel ratio.

Various devices have been invented to solve this problem, but what I present to you here is the simplest, most efficient and most applicable to carburetors today. It is the easiest to make, but the most difficult to adjust.

How does this device work?

With the throttle open, the depression sucks in air, plus an air-fuel emulsion coming from the emulsion well. In inaction the fuel level in the emulsion well is determined by the level in the float chamber. In action, air is sucked in through the air nozzle called an "air flow automation nozzle" and mixes with the fuel whose flow is limited by the respective nozzle.

The emulsion obtained in the emulsion well has a height in the emulsion tube greater or less, varying depending on dilution (depression). We would think that the proportion of sucked fuel will increase rapidly - nothing like that, because the air sucked harder and harder, will cause a progressive drop in fuel levels in the emulsion well, which in turn leads to an emulsion increasingly poor in fuel and the rate of this depletion will depend on the cross section of the two nozzles, on the holes in the pipe, on the cross section of the emulsion well, etc.

IT IS VERY IMPORTANT TO UNDERSTAND THE PRINCIPLE OF ACTION OF THIS DEVICE, WHICH IS THE SOUL OF THE CARBURETOR

It is pointless to "tinker" in the carburetor without this necessary knowledge. Thanks to this device we can get, with search and chance, quite acceptable result. You have already understood that in order to properly power an engine in any mode of operation, the carburetor must deliver a well-proportioned mixture at all times. Of course, perfection cannot be achieved, but every carburetor manufacturer makes every effort and uses different means to get closer to it.

In addition to the diameter of the diffuser, which must be sufficient so as not to impede the flow of air at full engine operation, the to carefully adjust the various elements of this device. Here's what to do:

- the exact choice of the fuel (main) nozzleLe bon choix du gicleur de carburant;
- the correct choice of the air nozzle;
- the correct choice of the emulsion tube;
- the correct choice of the cross section of the emulsion well;
- the correct choice of the fuel level in the chamber.

The parameters we need to regulate are numerous and it is difficult to achieve a good carburetion, but you can find ideas from one finished carburetor and the displacement of the engine to which it belonged.

Also know that there are different configurations of the emulsion well and its elements. For some carburetors, the nozzle is installed at the end of the emulsion tube (sprinkler) .. This is very practical for the various adjustments, as we can easily pull out the tube and its two nozzles with a simple screwdriver, without the need to empty the carburetor (see diagram).

In other carburetors, the emulsion well and its tube (sprinkler) are installed in the center of the diffuser - we can easily change the air nozzle and the emulsion tube (sprinkler), but access to the main nozzle is difficult.

Why we need to install a carburetor to an engine equipped with a PANTONE reactor.

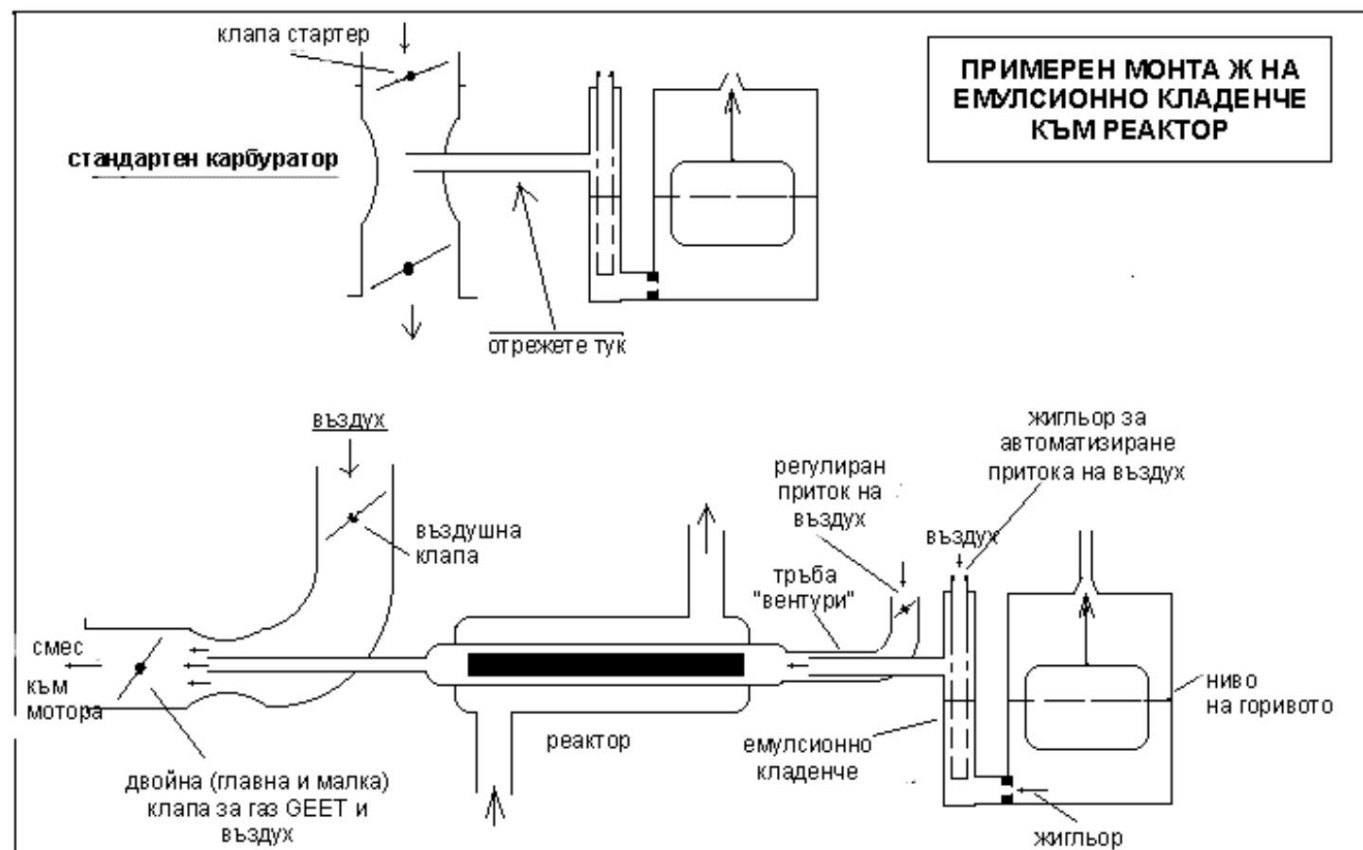
In the case of a carburettor engine, the addition of air to the emulsion takes place in the diffuser immediately at the outlet of the emulsion well.

In the case of an engine equipped with a PANTONE evaporator reactor, the reactor is located between the air source and the evaporator.

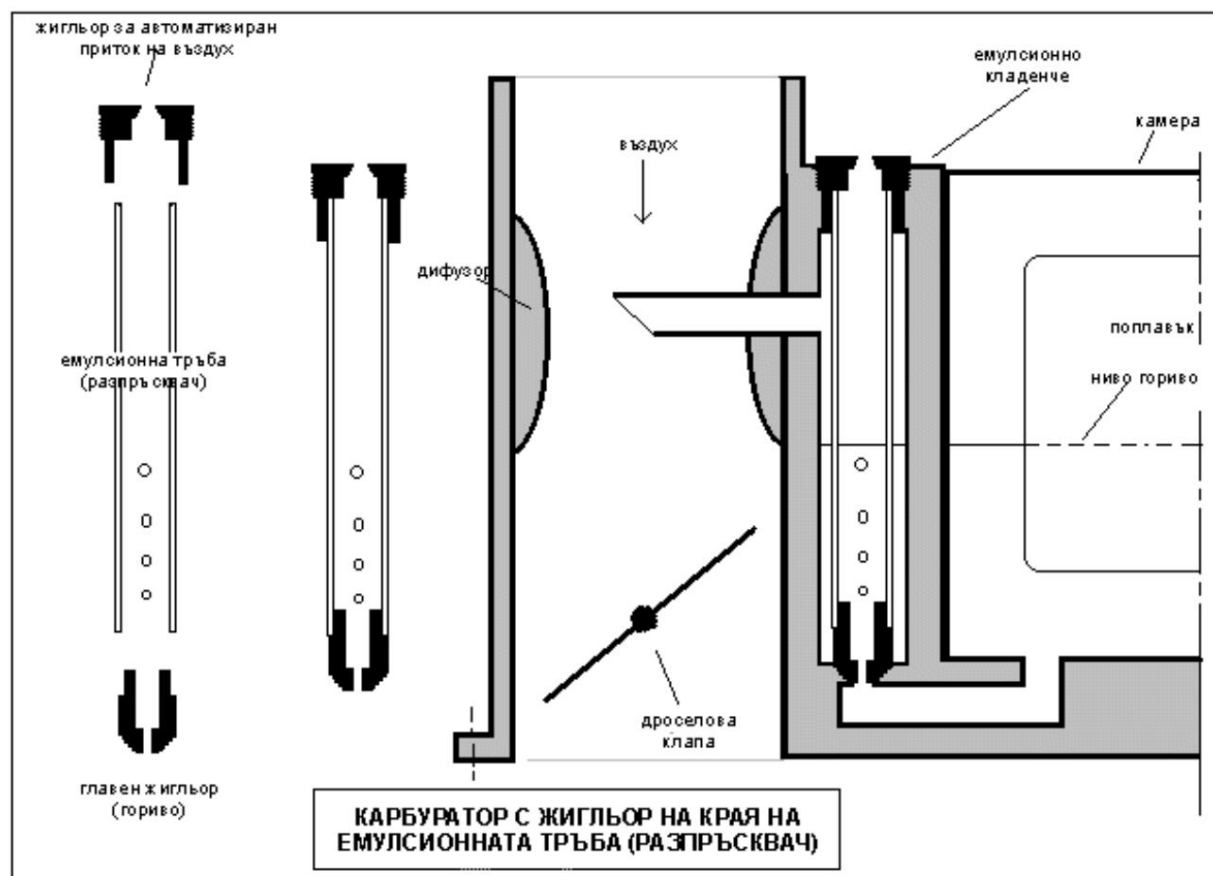
The emulsion well in the carburetor is something like a small evaporator, so we can decide to place the reactor between the air inflow and the emulsion well. The reactor will be used to transform the aerosol into a gas before adding air.

So the emulsion well is a kind of "mini-evaporator" which has the advantage of needing some air and especially a little dilution (depression) to work. Exactly what we need, because the discharge in the reactor is low due to the main source of air, which causes us so many problems and because almost always, according to different cases, you need to add a little extra air at the outlet of the emulsion well, this does not arrange things too much. This operation is performed by means of a long tube "venturi" and a small air valve added to the main (see diagram).

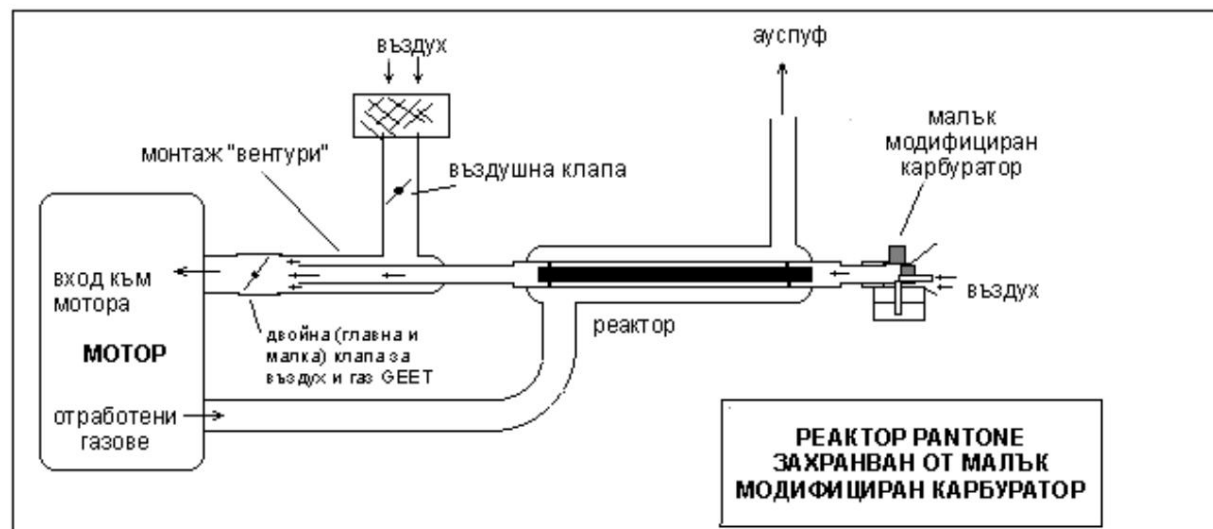
The diagrams below are much clearer than the long explanations, so you have no choice but to guess the exact adjustments of the emulsion well, which will be easy once you know what it is for and how it works.



I told you in advance that in some carburetors the emulsion tube (sprinkler) carries both the air and the fuel nozzle. This one configuration is interesting because we can quickly disassemble what is needed to change the settings without having to empty carburetor. With this type of carburetors we can decide to remove with a hacksaw the part that carries the diffuser and the throttle valve in order to be able to fix a pipe at the outlet of the emulsion well, which will be connected to the reactor (see diagram).



In any case, you will have to add a little air to the emulsion coming out of the well. This additional influx the air can be constant or adjustable (see diagrams).



Note: if you have, like me, machines and tools available, you can design an emulsion well according to your desire and use only the carburetor chamber, supplying here and there with nozzles or make them yourself.

It should be noted that after preheating the reactor for a few minutes with gasoline, all engines can be powered by gas oil. It is clear that at this point the engines are actually running on gas, with all the advantages that this brings.

If you decide to supply the engine with gas oil, you must keep in mind the adjustment of the carburetion parameters, which will not be the same as with gasoline, as long as the preheating of the reactor is done with gasoline, even with improper adjustment.

It is true that the operation of carburetors deserves more explanation, a whole book will not suffice, but what I have set out here, simplifying, should help you see more clearly.

EQUIPMENT OF PANTONE REACTOR WITH INJECTION DEVICE

Injection spraying is interesting because it can be easily controlled and adjusted for all engine operating modes. In addition, without the need for dilution (depression) or air to obtain it, its quality depends only on the capabilities of the injection, which is of the electromagnetic type used in automobile engines. Finally, it allows it to be removed one of the two air valves at the engine inlet.

Obviously, the injection is the dream solution.

The main inconvenience will be to be profitable with a 2.5 bar electric pump, injection and electronic pulse generator. The pulse generator can be made at little cost, but the pump and one or two injections can be found cheapest on the auto morgue.

Installation and especially adjustment is infinitely easier than the installation of carburetors, whose difficulties I have already described.

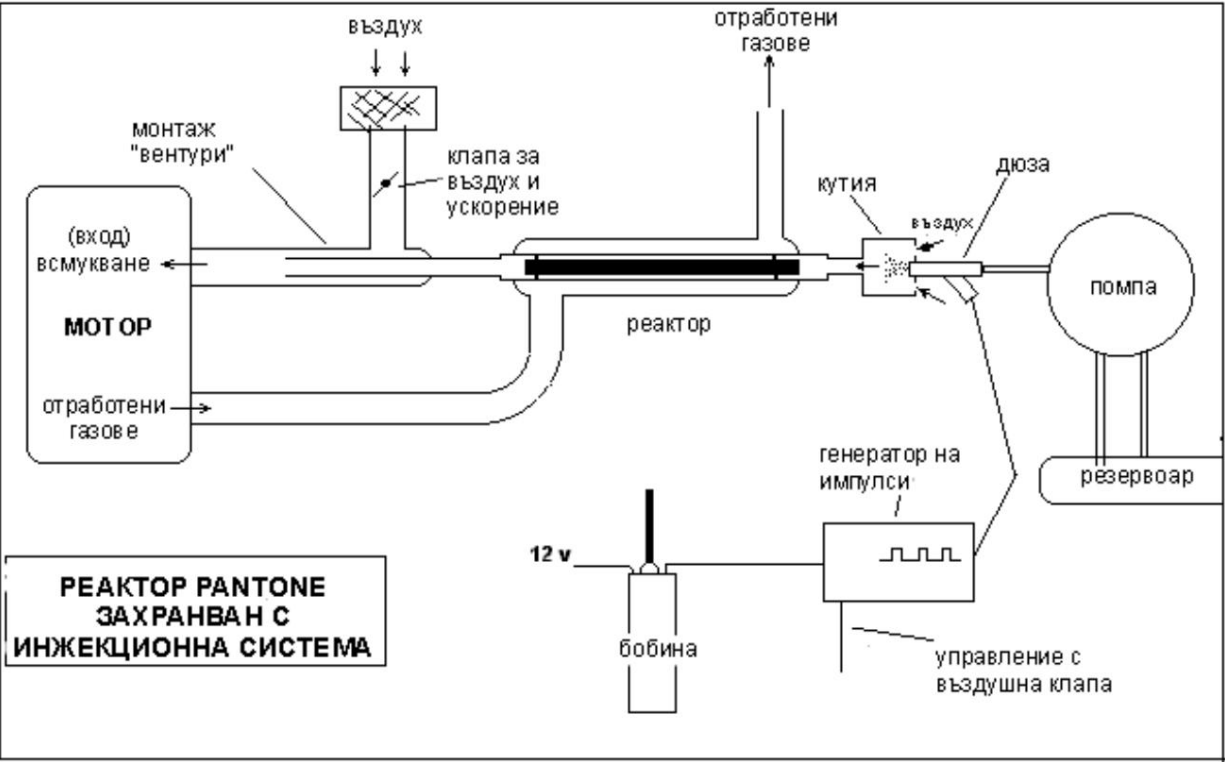
We see in the diagram that the injection nozzle is mounted on a small box, it itself is installed at the inlet of the reactor. On one side of the box must have small holes that can be partially closed. It is possible that these openings for the necessary air are operated by means of a valve (shutter) in some cases.

It should be noted that the size and shape of the box, the air holes and the location of the injection nozzle can be objects of guessing and searching for optimal parameters. But first, do everything in the easiest way.

Note: In carburetor installations, it is necessary to have two air valves to control the operating mode of the engine under the best conditions. In fact, one of the external access valves controls only the entry of air, and the other between the reactor outlet and engine inlet - controls the entire amount of gas / air that enters the engine. These two valves, together with a small gap, allow better dilution (depression) in the low-reactor. In injection installation, only the air valve is needed, as it is connected to the pulse generator, where it controls the pulse rhythm (ie the time between two pulses). But there is a risk of mild loss on the brake engine (adaptation to the engine of some trucks to reduce their speed) when we remove quite the gas, unless we close the air holes in the nozzle box, which is very good solution to prevent the reactor from cooling off most of the air entering the deceleration.

After all, everyone can choose the solution that seems to work best for the engine they want to equip. We can choose different solutions depending on whether the engine is running at constant power (electric motor) or is the engine of a vehicle.

I found that all gasoline engines run on gas oil after 2-3 minutes of heating the reactor, both carburetor and injection system, and with a ridiculously small amount of harmful exhaust gases compared to a traditional engine.



PERSONAL REFLECTIONS ON THE THEORY OF THE PRINCIPLE OF ACTION OF PANTONE REACTOR

[GO TO HOME PAGE](#)

What exactly is happening in the PANTONE reactor is the subject of discussions and debates.

As for me, I think the whole thing is based on the fact that the money of fuel that we force to pass in the narrow space between the shaft and the tube, is electrified by friction in this axis and in the walls of the tube. This electrification is necessary to obtain, otherwise the reactor will remain the most common means of heat exchange in order to increase the evaporation that has been experimented in the past and which only allows to slightly improve the air / fuel vapor mixture and therefore combustion.

In return, if fuel vapor is electrified, it is decomposed into fuel gas with characteristics quite different from those of ordinary fuel money. That is, we get this gas when the necessary conditions are met in the construction of the reactor and most of all when it is powered. The composition of this gas varies depending on the type of fuel that enters the reactor and it is always based on molecules of hydrogen, oxygen, carbon, etc.

In fact, Mr. PANTONE designed his reactor simply as a two-way heat exchanger. Exhaust gases circulate in one direction and heats the reactor, and the fuel vapor circulates in the opposite direction in the narrow space between the shaft and the pipe. The area, c which circulates fuel vapor is quite long, about 30 cm. In the first centimeters, in contact with the hot walls of the pipe the pulverized fuel is converted into "dry" steam, then the microdroplets of this "dry" steam continuing in the narrow space between axle and tube are electrified by friction, which becomes easy if they circulate at a sufficiently high speed and if the tube and shaft are not moistened with fuel.

It should be noted that all these microdroplets and molecules, just charged with static electricity, will be discharged immediately at approaching the walls (pipe or shaft) or each other. They attract, repel, charge, dilute, etc.

What is important to understand is that with each charge-discharge a small spark is produced, which causes the microdroplets to break up. to molecules, even to atoms.

All these sparks produce the familiar increase in temperature that we observe at the reactor outlet - a temperature that is higher than that of the exhaust gases.

This is a simplified illustration of the phenomenon, which is certainly much more complex and varies according to the products used. Yes that, it can be assumed that in some cases can occur a pre-combustion in the reactor to produce a product other than that produced by electrostatic shredding.

We now understand how important it is to obtain and promote the electrostatic process described above as much as possible.

If the fuel is not sufficiently finely sprayed before entering the reactor, or the reactor is not heated enough, its interior will get wet and get too little electrostatic charge. For this reason, the reactor will not produce the desired result and the engine will work in the traditional way, as it will not be powered by this special gas that the reactor is supposed to produce.

Note that if you supply your reactor with an evaporator and not a carburetor or injection, the charge will be very fast. at start-up as soon as the reactor starts to heat up, which prevents too much condensation and as a consequence wetting the reactor from the beginning because the fuel is much more dilute by evaporation and has a lower risk of condensation. For this reason, experimental assemblies using an evaporator work very well with volatile (easily volatile) fuels, even with the most basic design.

It is important to know the following conditions in order to make Mr. PANTONE's invention to its full effect.

prepare the fuel by evaporating it or spraying it as finely as possible when introducing it into the reactor;

heat the reactor so that fuel vapors do not condense inside in contact with the shaft and the inside of the pipe; in case of condensation, the inside of the reactor will be humidified and no electrification will occur;

you must make sure that the charge (static electricity that is charged in the fuel vapor and the inner walls of the reactor tube) be as intensive as possible; to achieve this, the circulation rate inside the reactor must be as high as possible in all engine operating modes;

Obviously, the first and third conditions are the most difficult to perform when the engine is running in low mode. I wrote about this at length in the previous ones pages.

WHAT IS THE BENEFIT OF THE "ELECTRICITY" PHENOMENON IN THE FUNCTIONING OF THE PANTONE REACTOR?

Let's see what happens to a single microdroplet of hydrocarbon or water circulating at high speed in a narrow space between the shaft and the reactor tube.

Given the length of the reactor, there will come a time when this microdroplet will "rub" against the axis or the inner wall of the reactor. the pipe. If the above three conditions are met, this microdroplet will receive a negative or positive electric charge opposite to this on the shaft or tube. As soon as it is received, this charge will change its polarity, either from a new contact with the axis or the tube to which it belongs

attracted because of opposite polarity, either from contact with an adjacent microdroplet, which is currently also of opposite polarity.

As I said, all these charges and discharges produce micro-sparks, which raise the temperature of the reactor more than that obtained from the exhaust gases.

Sparks cause the microdroplets to break down to the complete decomposition of molecules, even atoms, and all this in a situation of pre-combustion with all that may result.

I think that cracking chemists could give you more clarification based on to the relevant accurate calculations.

We now know enough to work on the PANTONE reactor and determine its dimensions and shape according to its application and its purpose. Personally, I designed reactors of glass, copper, Teflon - depending on the system of the pipe and the inner axis - as well as others, perfectly flat.

We must also not forget that the electric charge is obtained more or less easily depending on the type of droplets and the type of the inner walls of the reactor.

For example, if we have droplets of hydrocarbon fuel, the electric charge will be easily obtained, because we are dealing with a product that is very insulating, even at lower speeds, even in contact with conductive material such as steel. When we have water vapor, the electric charge of the molecules by rubbing in the steel becomes more difficult, because we have two materials with relatively good conductivity.

I also noticed that the performance of an engine equipped with a PANTONE processor improves over time. I think it's due to the fact that the inner walls of the reactor tubes and the surface of the shaft are covered little by little with a thin layer of insulating material, which favors the production of electric charges.

Now you know the three main criteria for the proper functioning of a PANTONE reactor, namely the need for high speed of gases in the reactor, which in some cases leads to the need to install a device (rotary valve) for periodically short obturation of the air inlet at the reactor outlet and directly at the inlet to the engine (suction) ([see page 5](#)).

With this device, after preheating, all gasoline engines work perfectly with fuel oil, which means that they become gas engines.

To solve the problem of gas velocity, it would be interesting to design a variable size reactor that shrinks at low operating mode and increases with increasing engine speed.

It would also be an interesting decision to remove the air inlet, which causes us so many problems, and to let the air through the varying reactor together with the fuel.

I am currently experimenting with this type of reactor and will let you know if the result is good.

DAVID M.

Since 1840, it has been known how to produce static electricity with steam, as evidenced by the following excerpts from "A Small Treatise on physics" by MJ Jamin, Gauthier-Villars Paris - 1882.

"William Georges Armstrong, English inventor (1810-1871) is known for his constructions of machines and cannons. His system cannons found application throughout the English army in 1858, making it famous and rich. In addition to cannons and hydroelectric machine, he invented various machines with hydraulic presses. "

Machine Armstrong. — Un mécanicien anglais, occupé à réparer des fuites dans la chaudière d'une machine à vapeur, vit des étincelles et éprouva des secousses quand il tenait une de ses mains dans le jet de vapeur et qu'il touchait avec l'autre la soupape de sûreté. M. Armstrong, qui étudia ce phénomène, reconnut que la séparation des deux fluides se fait, non dans l'intérieur de l'appareil pendant l'ébullition, mais au point où la vapeur sort en frottant contre les parois du robinet. M. Faraday prouva ensuite que la vapeur ne donne point d'électricité quand elle est surechauffée et sèche, tandis qu'elle en produit abondamment quand on la fait passer, avant sa sortie, dans une boîte contenant de l'étoffe mouillée, où elle se charge de gouttelettes liquides. Ce sont ces gouttes qui s'électrifient en frottant contre les parois du bec.

Quand l'eau de la chaudière est distillée, le bec devient négatif et le jet de vapeur positif. L'eau ordinaire ou saline ne produit rien. Avec des essences ou des corps gras, la vapeur est négative et la chaudière positive. La production de l'électricité dépend aussi de la matière des becs; elle est nulle s'ils sont en ivoire, et abondante quand ils sont métalliques; le bois de buis donne les résultats les plus énergiques.

La machine Armstrong (fig. 221) est fondée sur ces propriétés. Elle se compose d'une chaudière remplie d'eau distillée qu'on porte à 6 atmosphères. La vapeur s'échappe à travers des becs C après avoir été refroidie dans une boîte pleine d'eau D, qui est représentée à part dans la fig. 222;

17

Machine Armstrong - an English mechanic who fixes leaks in the boiler one sees sparks and feels the steam engine jet and with the other hand touches the safety valve. Mr Armstrong who studied this phenomenon, admitted that the separation of the two fluids does not take place inside the device during boiling, but at the point where the steam comes out, rubbing against the walls of the tap. Mr Faraday later proved that steam does not produce electricity when it does, but it is very hot and dry, abundant when we spend it, before leaving, through a box with moist tow, where it is stored in liquid droplets. It is these droplets that are electrified by rubbing against the walls of the outlet. When the water in the boiler is distilled,

en sortant elle contourne une lame verticale de buis sur laquelle elle frotte

Fig. 221.

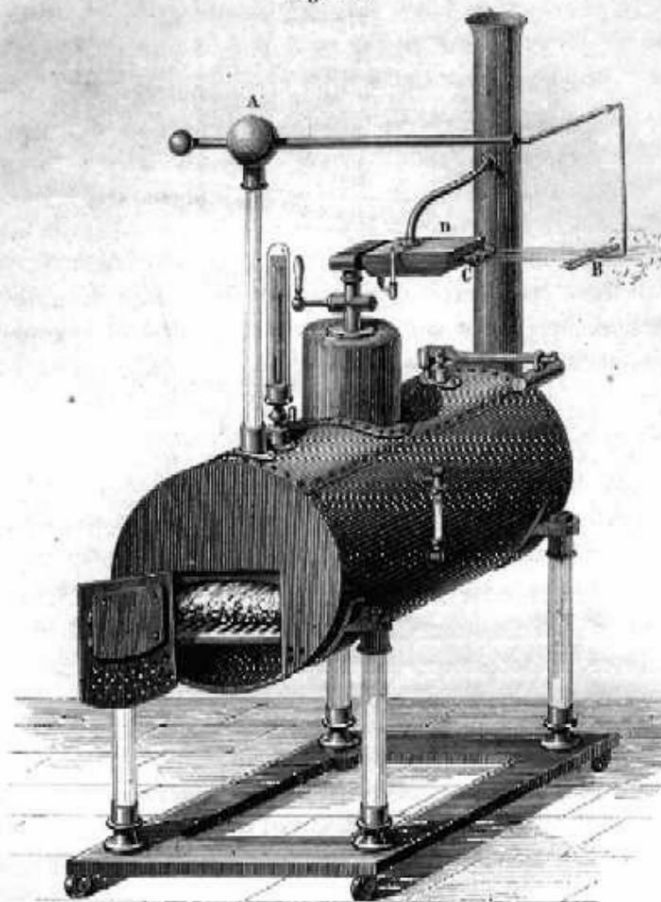


Fig. 222.

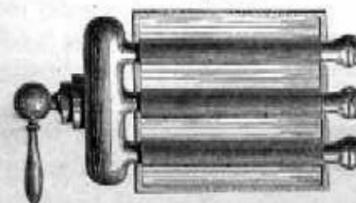
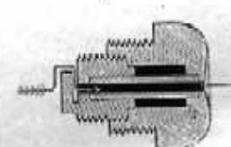


Fig. 223.



(fig. 223) et qu'elle charge négativement en prenant elle-même l'électricité

the mouth is negatively charged and the steam is positively charged. Plain or salt water produces nothing. Steam is negatively charged with oils and fats and the boiler is positively charged. The production of electricity depends on the material from which the outlet is made hole is made of ivory in which it is insulated. This machine produces a lot of energy. This abundant if it is made of metal. The boxwood tree, which is owned by the Polytechnic Institute in London and which gives the most energetic results. has a length of 2 meters, gives 46 jets, it produces sparks with a steady rhythm. of these qualities. It consists of a boiler filled with distilled water with a pressure of 6 atmospheres. The steam is passed out through the openings C after it has been cooled in a box of separate windings.	positive. On recueille cette électricité sur un peigne de laiton B qui est isolé. Cette machine atteint une grande énergie. Celle que possède l'Institution Polytechnique de Londres et qui a 2 mètres de longueur, donne 46 jets; elle produit des étincelles de 60 centimètres de longueur, qui éclatent d'une manière à peu près continue. positive. We collect this electricity on a bronze comb, which is zero if the metal. The boxwood tree, which is owned by the Polytechnic Institute in London and which gives the most energetic results. has a length of 2 meters, gives 46 jets, it produces sparks with a steady rhythm.
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AIMANTATION PAR L'ÉLECTRICITÉ. — Les procédés dont nous venons de parler sont complètement tombés en désuétude. On em-

ploie aujourd'hui presque exclusivement dans les ateliers les deux méthodes suivantes, basées sur des principes que nous développerons ci-après :

1° On se sert d'un *électro-aimant* fixe (fig. 67), sur les pôles duquel on fait glisser, alternativement et en sens contraire, le barreau à aimanter. L'effet de cette friction est de développer à l'extrémité qui abandonne l'aimant un pôle de nom contraire à celui qui agit, d'où l'on voit que les frictions inversées sur les deux pôles tendent à produire une aimantation dans le même sens.



Fig. 67.
AIMANTATION PAR UN COUCHANT.

2° Quand on veut produire une aimantation assez énergique, l'électro-aimant doit être puissant, et dès lors l'adhérence du barreau est telle, que l'opération devient incommode. D'ailleurs le barreau peut se trouver fortement rayé par la friction. On préfère dans ce cas faire mouvoir le long du barreau une bobine (fig. 68) traversée par un courant.

C'est Arago (1) le premier, après les belles découvertes d'Ørsted, qui

Magnetization through electricity - The methods we talked about have fallen into oblivion. Today, the following two methods are used exclusively in the workshops, based on the principles we will develop a little below: 1. We use a fixed electromagnet (Fig. 67) on the poles of which we slide sequentially and in the opposite direction, the magnetizing rod. The effect of this friction is to create at the end, moving away from the magnet, a pole opposite to the one acting, where we see that the reverse friction on both poles tends to create magnetization in

the same direction.

2. When we want to produce a very energetic magnetization, the electromagnet must be strong - in this case it attracts the rod so much that it becomes inconvenient to manipulate. By the way, the stick can be strongly scratched by rubbing. In this case we are satisfied to move along the length of the rod a coil (Fig. 68) through which

current passes.

First Arago, after the wonderful discoveries of Ørsted, who ... (follows below)

découvrit que l'on pouvait ainsi faire des aimants puissants, en soumettant l'acier et le fer à l'action d'un courant électrique. Une expérience intéressante a été faite en 1875, dans le même ordre d'idées, à l'Académie des sciences, par M. Thomasi. Ce physicien produit des aimants, non plus en soumettant du fer ou de l'acier, comme Arago, à l'action d'une hélice dans laquelle passe un courant électrique, mais en plaçant un barreau dans les spires d'un tube creux au sein duquel circule un jet de vapeur. Un tube de cuivre de 2 à 3 millimètres de diamètre est enroulé en hélice autour d'un cylindre de fer; on fait passer de la vapeur à 5 ou 6 atmosphères dans le tube, et le morceau de fer s'aimante et conserve son aimantation pendant toute la durée du passage de la vapeur.

Quelle est la cause du phénomène? M. Thomasi s'en est tenu aux faits, sans insister sur la théorie. Quelques physiciens ont avancé que la vapeur, en circulant, créait un courant thermo-électrique par suite de la différence de température entre la paroi froide extérieure et la paroi chaude intérieure du tube. Il semble que, s'il en était ainsi, l'égalité de température tendant à s'établir, le courant disparaîtrait vite. Il est plus probable que le frottement des gouttelettes de vapeur d'eau, entraînées contre les parois du tube, crée de l'électricité qui aimante les barreaux.

... found that it could be done that way
powerful magnets, subjecting iron and
steel electric current One

An interesting experiment was made in 1875
the same order of ideas in the Scientific Academy by Mr.
Thomas. This physicist created magnets without
subject iron or steel, as Arago, to
the impact of the spiral through which it passes
electricity, but putting a stick in
the inside of the pipe through which it flows
steam. A copper pipe with a diameter of 2-3 mm is
wraps spirally around an iron pipe; we release
steam with a pressure of 5-6 atmospheres through
tube, the iron cylinder is
magnetizes and retains its magnetism
cut all the time while passing through it
pair.

What is the cause of this phenomenon! Mr.
Thomasi sticks to the facts without
deepens in theory. Several physicists
suggested that steam, circulating, creates
thermo-electric current due to the difference
in temperatures between the outer wall, which
is cooler, and the inner heated
surface. It seems that if it really is
so, pi temperature balancing
the electricity will go out quickly. By that the
it is probably . friction between the water
droplets in the steam, and on the walls of the tube,
creates electricity that magnetizes
the stick.

PETROL ENGINE OPERATING WITH OIL HEATING OIL PANTONE OBTURATOR AND REACTOR

In pages 5 and 10, I described a successful air obturator.

With this device we can put an end to the problems with dilution (depression) necessary for the normal operation of the reactor, especially in the case when it is powered with oil fuel such as. fuel oil by means of a carburetor spray device.

In fact, as I have often said in my pages, everything that goes through the reactor must always be with enough required speed in any engine operating mode (see page 13).

Otherwise, in many installations, the air inlet located between the reactor outlet and the inlet to the engine causes unsolvable problems with carburetor reactors. This is normal, as this hole favors the inflow of air, which prevents the maintenance of the necessary depression in the reactor.

Therefore, on pages 5 and 10 I proposed a system with rotary valves that periodically close the air flow for a short time. through the suction stroke, so that during clogging the engine sucks only from the reactor, which allows it to operate with a carburetor in "starter" position.

**It should be noted that only with this system with a rotary valve I managed to
make gasoline engines work in good conditions with heating oil, with clean
alcohol or mixed with 40% water.**

Although the concept is simple, making the rotary shut-off valve described on page 10 would be a problem for those who do not have the necessary machine tools. Another disadvantage is that a large number of small motors running in generator sets or other appliances do not offer access to the drive shaft so that we can install a gear for the transmission belt of this rotary valve.

For my small 2 kW Honda generator set, I came up with a device that doesn't need to be connected to the drive shaft.

The device consists of a perforated disk rotating in front of the air inlet, driven by a small electric motor removed from a cassette player, turntable or other similar device.

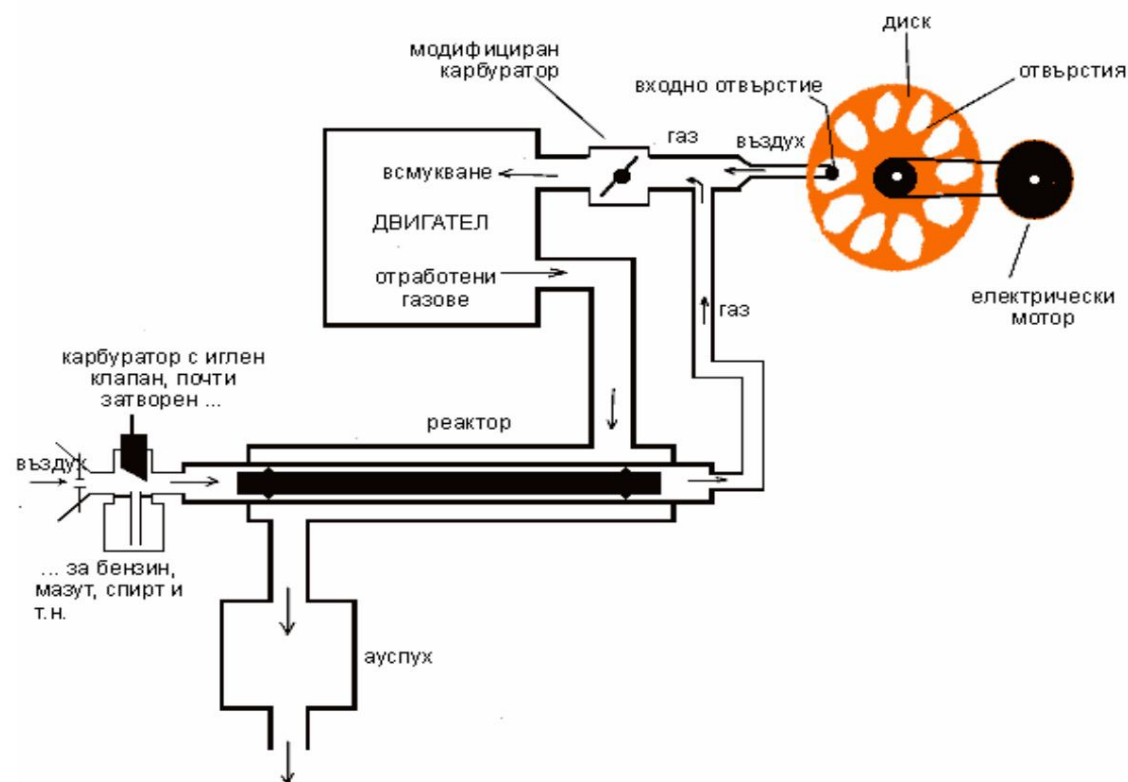
So this is a real installation, which can be done without special tools, but is extremely efficient.

This installation operates at a constant speed, regardless of the speed of the main engine.

The speed of rotation of the disk is about 4000 rpm in my case. Depending on the number of holes in the disk we get Numerous closures and openings of the air inlet during the intake stroke, even when the engine is running at 3000 rpm which is a normal speed for this small generator set.

For simplicity I used ball bearings and a disk from a damaged computer hard disk, the rest of the material is a small PVC board and synthetic resin, kneaded by hand, which is used for gluing and making various shapes.

See the photos below. An important advantage is that the engine speed is controlled by only one valve, which can be combined with the control lever of the unit.



If you manage to design this rotary air shut-off valve or similar, you will get surprising results and many problems you had with the Pantone reactor will disappear - this will be a big improvement in your work.

Note: Obviously, most of the devices I present on these pages are only available to experimenters, more or less equipped with tools. These devices allow the Pantone reactor to be used without changing anything in the engine mechanics. We can imagine that it would be possible to get better results in many cases if the engines were originally made

adapted for use with the Pantone reactor. For example, in order to avoid the need for a rotary shut-off valve, an additional suction valve could initially be provided. reactor valve specifically for an air compressor to facilitate the flow of air into the engine. In short, from the moment we master the operation of the reactor, we open the door to all possibilities.



General view of the small Honda unit 2, 2 kW



Top view: we can clearly distinguish the reactor, carburetor and exhaust. This reactor is made of an axis with a diameter of 8 mm and a length of 16 cm in a tube with an inner diameter of 10 mm.



Here we distinguish the modified carburetor, in which we use only the throttle valve. The inlet (suction) of the reactor and the air vent are on the same side as the throttle valve (see diagram). Note: Do not allow any holes in the nozzle or elsewhere to avoid parasitic airflow.



Carburetor inlet view in starter position



Three views of the small electric motor driving the rotary shut-off valve. Notice the belt pulleys and the rubber transmission belt.



The motor can be powered by battery, mains, etc.



We can clearly see a hole in the disk just in front of the air hole.



Here the air vent is clogged. It is cut in the PVC plate, screwed on the inlet of the modified carburetor, in place of the air filter. This carburetor only serves as a mixer where Pantone gas and air are mixed, it also serves as an accelerator valve.



The flexible silicone tube that connects the reactor outlet to the modified carburetor in which we used only the throttle valve to control the engine speed.



Here we see the handle fixed to the axis of the throttle valve of the modified carburetor.

M.David Xavier Faure: photos and scheme

GENERAL INFORMATION - WATER BURNING

After months of silence, during which I had the pleasure to find that there are many who are interested in the invention of Mr. Pantone, I take the pen again. I am eloquent, but I know that thousands of people around the world are passionate about this topic.

I also follow the discussions in the PMC-France forum, and this encouraged me to make this new page to highlight some obvious truths about fuels, motorcycles, etc.

I think that working on a topic, it is good to take stock from time to time about the established scientific knowledge on the given topic. Thus, by comparison, we can make some progress in this work.

GENERAL KNOWLEDGE OF ENERGY, ENGINE AND POWER

An engine, petrol or diesel, burns in any case a mixture of fuel / substance that supports combustion - that is, any fuel mixed with air that contains oxygen, which in turn supports combustion.

However, one liter of this fuel / air mixture will potentially contain more or less energy depending on the choice of fuel.

When dealing with motorcycles, we must always keep this in mind in order to correctly interpret the results observed with different fuels. mixtures that we experiment with, and not to compare what is not compared.

HERE'S HOW MUCH ENERGY GIVES 1 LITER OF FUEL MIXTURE CONSISTING OF DIFFERENT FUELS at an ambient temperature of 10 to 15°C

MIXTURE	contains (calories)	real (calories)
air + sprayed gasoline	0,768	0,720
air + butane		0,843
air + propane		0,845
air + methane		0,760
air + coal gas		0,715

air + acetylene		0,990
air + carbon monoxide		0,870
air + hydrogen		0,880
air + gas obtained from a gas generator		0,590
air + alcohol	0,800	0,760
air + gas oil		0,600
air + fuel oil (the one used for a boiler in the center. heating)		0,570
air + gas PANTONE	?	?

These figures are theoretical, they correspond to a complete combustion of the fuel in an ideal air-fuel mixture.

In practice, the energy qualities depend on the fineness of the spray - if it is coarser, there will be a loss due to incomplete combustion.

This plate is a good basis for reflection, it allows you to calculate approximately (50% for a classic motorcycle) energy loss through the exhaust, which will then be useful to calculate the possible "water consumption" with Mr Pantone's device.

Now that we have the numbers in front of us, we will be able to start thinking positively, but without forgetting that there are no miracles in the field of energy.

Examining the table, we first see how much the power of the same engine designed to run on gasoline can vary, if it we feed with different mixtures of different fuels (except for fuel oil, provided that it passes through the Pantone reactor).

This table can help us learn more about this gas mixture coming out of the reactor with a composition that probably varies according to the fuel we use.

We know too little about this gas mixture. However, on [page 8](#) I present to you a small burner, my personal making, which can us learn a lot.

Some were intrigued - it's just a means to notice some features of this gas mixture and the difficulties we may encounter in reactor settings, spraying, etc.

OBSERVATIONS MADE ON MY BURNER "

WITH HEATING OIL

A LA PANTONE "WORKING

1. In cold ignition, what comes out of the reactor is only pulverized fuel oil, which burns normally with a yellow flame.
2. After about 2 minutes, when the reactor is already heated, what comes out burns with a blue flame 2-3 cm long, then the flame becomes more and more short, burns even inside the hole, almost transparent, with a characteristic burner sound.
3. At this point, a flat piece of steel (blade of a metal saw) placed in the flame, quickly turns white and releases a molten drop.
4. If we put out the flame, what comes out of the reactor looks like cigarette smoke when we laugh with the air. It should be noted that in 3-4 seconds this smoke grows in a space of 2-3 cubic meters.
5. The odor that spreads is that of gasoline or diesel mixture.
6. If we distill what comes out of the reactor, we get a yellowish product, a little less oily than the fuel oil that burns. like him in a small saucer (MD- carburant Makhonine?).

Note: Thanks to this small installation, I noticed that when we run hydrocarbon fuels through the reactor, it is better to heat only a third of its length, that of the exit, of course.

COMMENTS ON THESE OBSERVATIONS

Observations N ° 2 and N ° 3 allow us to claim that we are dealing with a product that can develop a temperature of 1400 ° C when mixed with the surrounding air as we manage to melt a piece of steel. Who is bidding? The speed of this flame is very high as the flame shortens and descends only into the burner hole in accordance with the rise in temperature and depending on the good regulation of carburetor. It also becomes almost transparent (see photos on [page 8](#)). _____

Who could achieve such a good result with ordinary fuel oil and air without the Pantone reactor? _____

For comparison, here are the possible temperatures in Celsius ° C with:

propane or butane + ambient air	950 °
propane or butane + pure oxygen	1400 °
hydrogen + ambient air - about 20	50 °
hydrogen + oxygen - about 2	50 °
Pantone gas + ambient air - about 1400	°
Pantone gas + pure oxygen	?

What temperature would we reach if the Pantone product or gas burned mixed with pure oxygen?

So the question is, how is it possible to have such a high flame temperature when combustion takes place in ambient air containing 80% nitrogen - a gas that generally interferes with combustion? ... *The opinions of chemists, specialists in the field of combustion of all kinds are good come ...*

It is good to talk about high temperatures, but what are the energy values of one liter of a mixture composed of Pantone gas and air in the right proportions? This is a question that needs to be answered in order to advance in Pantone technology, as if not it I know I can't put an exact figure in my comparison chart with other fuels. We need to get back to that point later and yes we act by comparison with hydrogen or its derivatives.

FUEL SAVINGS

Let's talk a little about the fuel economy that some people observe in their experiments.

When a petrol engine runs on petrol, the combustion is in any case incomplete, in different proportions depending on the quality of the combustion, injection, operating mode, etc. This causes 10-15% losses.

Therefore, the energy values of one liter of gasoline / air mixture (0.76) can easily fall to 0.70 and even lower, and to get the required power, increase the speed or enrich the mixture, in short, waste gasoline.

In contrast, if the same engine runs on other mixtures in the table, we can assume that the energy values given in the table will be fully used, as the gases mix very well with the air - so the mixture will burn completely without waste. , for which of course the mixture must be in the right proportions.

Now, if we equip this engine with a **Pantone reactor** powered by gasoline, in the same operating mode, the engine will necessarily consume less, with varying proportions, depending on the above.

This is obvious because the engine runs on gas and there is no sawdust caused by poor mixing with the air. Naturally, that's why the reactor it must be properly powered at all levels of engine operation, but that's another matter.

At first glance, it seems to us that the savings are realized only at this level.

Let's also not forget that the same motor, without modifying the compression level, will develop different powers depending on the energy values of the **fuel mixture**. That is, the same engine will develop more power with propane or butane than with gasoline. He it will be more powerful with gasoline than with solid fuel gas. It will obviously be more powerful with hydrogen or acetylene than with others gases. However, it should be noted in passing that it is not very appropriate to supply a gasoline engine with gas, because we can, without the risk of

self-ignition, to significantly increase the level of compression if its mechanics are solid enough, or to use a diesel engine that is designed to withstand high pressures.

In this way we use the maximum energy of the used gas, working with an efficiency of about 30 to 35%.

So it makes no sense to burn gas in a low compression engine with an efficiency of 25%.

It follows that if we properly install a Pantone reactor on a gasoline engine without modifying the compression level, we will have as a result:

- Gasoline savings
- Almost no air pollution
- Unchanged efficiency
- Indefinite power, close to the energy values of such fuel mixtures.

The same case, but with an increased level of motor compression:

- Gasoline savings
- Almost no pollution
- Increased efficiency
- Increased power

Note: It is clear that this engine, overcompressed or not, when equipped with a Pantone reactor, can run on any hydrocarbon fuel, mineral oil, vegetable oil and what not, provided that you manage to spray it finely and possibly heat it, for to pass it through the reactor.

In return, you will need to start your engine with gas (propane / butane) until the exhaust gas heats up the reactor to the required temperature.

ABOUT THE WATER

The results obtained with the Pantone water and reactor remain a real mystery. I will list the most reasonable explanations below.

BURNING WATER IN A MOTORCYCLE.

In this last part of my "word" I will tell you about the possibility of "burning" water in a motorbike thanks to Mr. Pantone's reactor. But before that, as a start, it would be appropriate to clarify what we know about the experiments made in this area.

All these experiments are similar and consist in introducing a small amount of water into the fuel mixture by spraying at the entrance of the engine before or after the hood.

The result, it seems, is a small increase in engine power, hence a slight decrease in fuel consumption by 2 to 5%. An increase in the amount of water injected has the opposite effect. The exact reason for this small increase is not entirely clear, we can suppose that:

- The water turns into steam and helps the pressure in the cylinder to last longer at the beginning of the lowering of the piston. Corollary at which the short-term pressure at the time of ignition (explosion) will be lower and will last a little longer (diesel engine).
- or the injected water decomposes in the process of combustion into oxygen and hydrogen, which burn back immediately water that turns into steam. As a result, this sequence of transformations leads to a longer duration of the moment the pressure at the time of ignition, as in the above case.

In these two cases it should be understood that the injection of water does not give additional energy to the engine, but in this way the engine has tends to run at a constant pressure rather than a constant volume, as with a diesel engine.

So it is quite possible that the observed slight increase in power is due to this "stretching" at the time of pressure.

CONCLUDED

- **first rule** - in order to hope to achieve something significant with water (with the energy it contains) we must we avoid subjecting it to any transformation, together with the calorific energy of the fuel mixture located in the inside of the combustion chamber and the cylinder.
- **second rule** - if we want to take advantage of the energy of water, we need all the transformations to which it is subjected - decomposition or evaporation - **to be partially** or fully realized with the **"lost"** energy of the exhaust gases and possibly on the water from the engine cooling system.

A little reminder of the decomposition of water at high temperatures

In some cases, it is assumed that the water decomposes at a temperature of about 900 ° C (for example, if we pass it through burning coals of the specified temperature, as in a gas generator).

Decomposition of one liter of water (1 kg) at 0 ° requires 3830 calories, or slightly less at normal temperature of environment.

So one calorie breaks down 0.265 grams of water.

One gram of decomposed water gives 1.24 liters of hydrogen and 0.63 liters of oxygen. One liter of hydrogen that burns with 0.63 liters of oxygen gives 3.09 calories. One liter of perfect mixture ($H_2 + O$) gives 2.03 calories.

The 3830 calories used to decompose one liter of water were eventually recovered from the combustion of the two gases produced by decomposition in the form of heat and steam. To

turn 1 gram of water into steam at $100^\circ C$ you need 0.637 calories. One gram of steam with a temperature of 100° represents 0.637 calories. One gram of steam with a temperature of 100° covers the volume of 1.29 liters one liter of air with a temperature of 100° represents 0.029 calories.

Note: so far there is no evidence that the water injected into the Pantone reactor actually decomposes, as is very likely. occurs with hydrocarbons, as the temperature is insufficient at best, except for the electrostatic (plasma) process in the reactor promotes decomposition or if the reactor functions as a miniature gas generator.

This suggests three possible possibilities:

1. **water decomposes completely in the reactor** - I personally do not think that water can decompose in the reactor when it passes through it alone (as in installations made with a diesel engine on a tractor or with other machines)
2. **when water and some hydrocarbon fuel feed the reactor, the water decomposes in contact with carbon at high temperature** - when only water mixed with air passes through the reactor, this air / steam mixture decomposes in the engine - Water may decompose if we assume that the reactor acts as a small gas generator, ie. when there is also hydrocarbon fuel that passes through it along with water. When a mixture of hydrocarbon fuel, water and air passes through the reactor, heated to a high temperature by means of the exhaust gases, we can assume that the water decomposes in contact with carbon. In this way, a gas mixture containing hydrogen, carbon monoxide and nitrogen leaves the reactor. Hydrocarbon fuel provides a portion of hydrogen. Water also provides hydrogen. Carbon monoxide comes from the carbon in hydrocarbon fuel. Nitrogen comes from the air that has passed through the reactor. This is a very likely possibility. 3. **when only water mixed with air passes through the reactor, this air / steam mixture decomposes in the engine** - the water

converts to dry steam, then decomposes in the engine, adding the plus of calories coming from the heated exhaust reactor. In particular, the steam entering the engine at a temperature of 100° to $300^\circ C$ will use less energy than burning fuel for its decomposition, which will lead to a fuel economy and a "snowball" effect in the amount of injected water depending on the increase in engine temperature and speed. It should be noted that this simple solution is applied to diesel engines of tractors, etc., which are not at all modified to be powered by gas oil, but only equipped with a Pantone reactor that treats water. **It is with this application of The PANTONE system has the most sensational results**, as it is very easy to adjust.

DETAILED CONSIDERATION OF THIS THIRD OPTION

I am sincerely convinced that the explanation that follows is the most logical, which explains the sensational results achieved by diesel motors equipped with a Pantone reactor.

Short preface:

If we traditionally inject a very small amount of water at a temperature of 10-20 ° C in the engine, it will decompose at high temperatures. of combustion of the fuel mixture, occupying a small part of heat energy and reducing theoretically for a very small part of time combustion temperature, i.e. pressure. In fact, the decomposition product, which is a perfect mixture of hydrogen and oxygen, will explode per second and will restore the energy occupied by the fuel in the form of calories, increasing the temperature, respectively the pressure of the gas volume with the help of steam, appeared during the combustion of hydrogen and oxygen. I have already explained the small profit we get from this way.

Let us now consider the case of a diesel engine on a tractor equipped with a Pantone reactor through which hot and humid air coming from evaporator or other device (see [page 7](#)) .

Without complicating matters, we can do a few simple calculations that speak for themselves.

These calculations will be made:

- based on one liter of air sucked in by the engine, as we use a diesel engine;
- for one injection of fuel oil, representing an initial 0.5 calories (80% of what is possible for 1 liter of air);
- for one injection of 0.05 grams of water;
- based on the fact that the loss of calories through the exhaust is estimated at 45% (because it is a diesel engine)

Exhaust losses 45% of 0.5	0.225 calories	
Other losses 20% of 0.5	0.100 calories	
Recovered mechanical energy 35% of 0.5	0.175 calories	
EVERYTHING:		0,500 calories

So, let me remind you that water decomposed with the help of calories occupied by burning the air / fuel oil mixture does not give much benefit, because the energy occupied by this combustion is returned in a second.

In return, if this water in the form of steam mixed with air produced some amount of energy passing through the reactor (usually lost calories) is injected into the engine, this amount of calories will not be deducted from the combustion of fuel oil injected into motor. It should be well understood that it is at this level that power increases and fuel savings occur.

So instead of embarking on complex calculations that will be inaccurate anyway, given the complexity of problems with combustion in engines, it is enough to calculate the amount of calories contained in water and air, which will penetrate the bike, and we will have approximately the level of realized savings. This amount of calories, or in other words the energy value, represents a possible increase in power or rather the possible economy.

Let's take a diesel engine that has reached its normal operating temperature, which swallows air with a temperature of about 20 ° C. After compression (40 kg) this air will reach a temperature of 700 ° C.

If this same diesel engine swallows a mixture of air / water vapor with $t = 100^{\circ}\text{C}$, after compression the temperature reached is theoretically significant, but in practice I think it holds at about 1200 ° C.

By the way, it is on this point that the help of gas and steam compression specialists will be welcome, as what happening in the bike at this point is not uninteresting.

Ultimately, fuel oil is injected into a combustion chamber, where the air and steam are at a temperature that is difficult to control. determine exactly, but it is significant.

No matter how high this temperature is, it can only improve, because fuel oil, after injection, can also decompose. well, as well as water vapor.

We understand immediately that what is very bad for a motor that operates normally becomes a necessity so that we can "burn water", ie. to heat at high temperature what enters the engine.

In this case the motor includes:

- 0.9 liters of air with $t = 100^{\circ}\text{C}$, or 0.020 calories;
- 0.05 grams (0.62 liters) of water vapor with $t = 100^{\circ}\text{C}$, or 0.032 calories;

ie a total of 0.052 calories.

We can immediately say what the approximate savings on the energy contained in the injected fuel will be.

This amount of energy taken from the exhaust is very small for the bottom such a small amount of exhaust water, 0.052 of 0.225.

Since we injected 0.50 mud. fuel oil, we can conclude that 0.052 cal. represents 10%. For the same energy supplied by the engine, consumption will be reduced by 10% and we only need to inject 0.448 cal of fuel into the engine.

I know that my way of calculating will terrify an engine specialist, I know that I have not included many parameters in the account, because I have no desire to write a whole book, but the end result has not suffered much. Even if the calculations are childishly simple, the most important thing is **to he knows what is happening in** order to know how to work when we move specifically to practice, and this has not been done so far.

Once you understand this, you can now inject more and more water vapor (4-5 times more), with a little air - the necessary the amount of fuel oil we inject to maintain engine power will decrease progressively, and at some point the fuel oil will only be as a "matchstick" that activates the process.

It has been observed that while maintaining a significant capacity, some tractors consume **4-5 times less** than in normal mode. There will always be enough energy in the exhaust, provided that calorie losses are avoided when the engine is running at low mode.

If we think about it, there comes a time when it is not the energy from the fuel injected that produces the mechanical power of the engine, but that recovered from the exhaust gases.

In fact, in our case (see below) the mechanical energy used is only 0.175 cal. Of the 0.50 calories invested and with small addition of water, 0.05 grams represents 0.032 cal.

OBSERVATIONS

On the issue of hot air entering the engine ..

Anyone who has not yet mastered what will be said will say that it is very bad to put hot air in the bike. In a classic bike, yes, but not in a motorcycle equipped with a Pantone reactor to burn water.

He will explain to you that the fuel injected while the engine is running at full mode will not have enough oxygen to burn, as expanded air contains less of it, and that the engine will not give its full power, and that the efficiency depends on the difference between the temperature of what enters the engine during intake, and what comes out of the exhaust.

In fact, with the Pantone system, we have to learn to think differently. With the Pantone reactor, in the cold, the diesel engine "swallows" enough air, which is needed in all modes, because the reactor is cold (it is possible to remove the possible injection of water). Little by little the reactor raises its temperature and we can start injecting water in very small quantities.

the more the engine increases its power, the more water we can inject into the reactor.

It should be noted that when the engine is hot enough, it runs on gas at 100% as even the injected fuel oil decomposes, igniting. At this point, the engine runs on gasoline, not diesel.

~~In conclusion, the more the engine "forces", the more water we can inject, the less fuel oil will be consumed and less oxygen is needed, which means that the inlet air temperature is not important at least at this level.~~

It should also be noted that at the thermal level, the operation of a motor with a Pantone reactor is completely "shifted", such as what enters the engine has a very high temperature - above 100 ° C - and the gases that come out are much lower than normal.

~~Precautions to be taken when installing the reactor to the exhaust pipe~~

In high operating mode, there is no problem to heat the reactor enough and heat the maximum amount of water. But if we want to we go from full mode to lower without risking too much cooling, we have to take care of the outlet of the exhaust pipe so that the gases do not cool too much in the cylinder head or in the exhaust duct. You need to put a thicker gasket than insulating material between the exhaust duct and the cylinder head to prevent the accumulated heat in the duct from rising to the cylinder head. This exhaust duct can be covered with insulating material as well as the reactor.

NOTE ON TOPIC REACTOR USED TO BURNER

It is not very interesting economically to make a burner for a central heating boiler with a Pantone reactor, unless we want to design a mini-boiler with a chimney similar to one with gas heating. There is no point in trying to "burn" water, because there is no lost energy to recover, or if there is, it is too little. In addition, the concept of a quality and safe burner is a very delicate thing, and the boiler must be made entirely of special for such a burner.

NOTE ON GAS TURBINE MOTOR WITH PANTONE REACTOR.

Given that the gas turbine engine has an interesting efficiency, we can imagine a turbine engine powered by a reactor that will decompose fuel oil or other petroleum products, vegetable waste, household waste, etc.

*I hope that this new page will help you in your work on the Pantone reactor, and will even help you discover new ones.
aspects of this system that cooks us even more surprises.
David M*

ENGINE IMPROVEMENT IDEAS

This page, which goes out of the context of PANTONE and which I define as a fun digression, addresses a particular issue concerning the strange way in which the expansion of burning gases in a petrol or diesel engine is exploited. That might be interesting possibly engine designers.

If we look at the diagram of the operation of such an engine, we noticed that at the end of the expansion stroke, just before the piston to reaches the bottom dead center, the exhaust valve opens at a time when there is still significant residual pressure in the cylinder. This pressure is about 25 bar per square cm for the diesel engine and 3 to 4 bar for the petrol engine.

This means that flammable gases are not given the opportunity to expand completely and thus a significant part of the the work done, which is disposed of cleanly and simply through the muffler.

In this way we lose at least 15% of the work done during fuel combustion.

It is obvious that if we tolerate this situation, it is because we cannot act reasonably otherwise unless we make engines that have a piston stroke that is longer during the expansion stroke than during the suction stroke. So it's out of the question present similar acrobatics, although some have already worked on the issue.

Personally, I think that something can be done to avoid this waste by approaching from a different angle, especially when it comes to diesel engines.

In practice, there is a growing tendency to equip diesel engines with turbochargers driven by exhaust gases. This allows these engines to be "trampled" with air, which artificially increases the working volume of the cylinder and gives an extra power.

Of course, this extra power has its energy cost, as the turbo, as well as the exhaust pipe, filter, etc., prevents the discharge of exhaust gases.

If we take a closer look, outside the moment of expansion of the gases (incomplete), everything that happens during the 4 cycles consists in wasting most of the energy produced by this expansion.

It should also be borne in mind that the turbocharger to the exhaust gives additional power only from a certain operating mode further, and this additional power is desired rather at low operating mode, as the result we get with mechanical volumetric compressor (pump type).

To avoid this inconvenience, the production of turbine compressors driven by an electric motor is envisaged. Energy however, the cost of such a device will be significant due to the low efficiency of electricity produced in a car.

Recalling all this, I recommend that you use the residual pressure at the end of the expansion stroke, which is usually lost, to turbo drive, it itself combined with a special alternator (alternator), and all this without the engine being slows down, thus using this 15% wasted energy that represents this residual pressure.

To achieve this result, the engine must be made with one outlet at the bottom of the cylinder and slow down the moment of opening the outlet to open just after the lower dead center of the piston at the end of the extension.

Here's what happens in a diesel engine modified this way

At the end of the expansion, the piston will open the small outlet a few degrees before the bottom dead center, just as the outlet valve needs to open.

Exhaust gases exit through the small exhaust and can drive the turbocharger turbine without slowing down the engine, as as the piston is downhill, moreover, all the resistance to the emission of these gases is utilized by the motor so far, in which the piston begins to rise.

Piston at bottom dead center, the pressure in the cylinder drops almost to the level of atmospheric pressure.

The piston starts to rise and the exhaust valve opens the moment the piston closes the small outlet. Must be noted in passing that the effort to open the valve is minimal due to the low pressure in the cylinder.

The remaining exhaust gases in the cylinder are expelled effortlessly through the valve when the piston is mounted. It should be noted that the cylinder head and the valve will be exposed to less heat, the valve may even be smaller.

Once the piston passes the top dead center, the intake starts, the turbo injects air into the cylinder without slowing down the engine, as as the turbo works "for free" and the air pressure helps a little to lower the piston.

When the piston reaches the bottom dead center, it detects the small outlet, some of the air exits through this hole, drawing with it part of the remaining exhaust gases.

After passing the lower dead center, the piston begins to rise and clog the small outlet.

The turbo continues to inject air into the cylinder as the piston rises a few degrees, then the exhaust valve closes.

The cycle continues normally until fuel injection.

Following these explanations, it is not difficult to understand the benefits of such a diesel engine design. The advantages are many, not all described here, and the work taken to install a second outlet in the cylinder is nothing compared to these advantages.

It should be noted that such an engine will have a more logical operation in terms of the pulsating mode (?) Of the exhaust gases or not, passing through it. In fact, most of the volume of these gases follows a more direct path as it passes through the motor, which is good for reasons related to resonant phenomena.

Given that today more and more cars are equipped with electrical appliances, it would be useful to produce energy that is usually lost to run the engine. With this solution it is possible to provide connected to the turbocharger is high-speed, and it is connected to a magnetic reducer. This alternator, producing current for the battery, will be able to serve as a motor and drive the turbocharger when the engine is running in low mode, which with a classic turbocharger is not possible and a great inconvenience.

Obviously, such an engine can also be equipped with a PANTONE reactor, which allows to "burn water", which I have outlined in previous pages, in particular on [page 15](#). _____

Next

Source

www.quanthomme.com and www.quanthomme.org